Board Diversity and Career Progression of Women: Evidence from SB 826*

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Abstract

Over the past two decades, many countries have implemented board gender quotas to increase female representation on corporate boards. By design, these mandates mechanically increase female representation at the executive level. What remains unclear is their broader impact on women in entry-level and mid management level roles. We examine these effects using California's 2018 board gender quota as a natural experiment and a novel employer–employee dataset. We find increased hiring and promotion, with no overall change in exit rates. Entry-level women experience increased hiring and improved retention, while mid-management shows increased hiring and internal advancement. To investigate potential mechanisms, we examine employee ratings from Glassdoor and find improvements in workplace culture, compensation and benefits, career opportunities, and work-life balance. The effects on hiring, promotions, and Glassdoor ratings are stronger in firms where female directors serve on board committees and have greater decision-making authority. Taken together, our findings suggest that female representation on boards can improve women's career opportunities within firms, particularly when female directors have the authority to influence firm policies.

Keywords: Board Diversity; Gender Representation; Corporate Governance; Board Quotas; Career Advancement; Labor outcomes; SB 826; Hiring; Promotion; Employee Turnover

JEL Codes: G30; G34; J16; J21; J24; J63; M14

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1 Introduction

Although women's workforce participation has expanded considerably, they remain underrepresented in executive and board-level positions. The 2017 U.S. Census found that only 27.6% of executive positions (U.S. Bureau of Labor Statistics, 2018) were held by women, even though women comprised 47.5% of the workforce. This gap in leadership representation is called the "glass ceiling," referring to subtle barriers that can limit women's advancement into top leadership roles.

In 2023, LeanIn.org and McKinsey & Company released their annual "Women in the Workplace" study, an in-depth exploration of women's experiences across 329 companies. They examined the challenges women face as they try to climb the corporate ladder and identified a major obstacle: the struggle to advance from entry-level roles to managerial positions. This initial promotion barrier limits women's representation in managerial roles and, in turn, hampers their advancement to higher levels of management within organizations (LeanIn, 2023).

Many countries—including Norway, France, Germany, Italy, Belgium, and Spain—have adopted board diversity initiatives such as gender quotas or disclosure requirements. In 2022, the EU mandated that listed firms have at least 33% women directors (Table A1). A policy that changes who sits on the board directly addresses representation at the very top, but whether it changes opportunities for women further down the organization still remains an open question.

Proponents of gender quotas argue that women in leadership positions can positively affect the career development of women in mid-management and entry-level positions (Smith, 2014). Women in leadership roles can help address unconscious bias (Beaman et al., 2009; Rao, 2019) and implement better hiring and promotion policies within the company. They may also serve as mentors and allies (Cai et al., 2022) to support career development.

On the other hand, critics argue that quotas may lead to tokenism, whereby women are

appointed primarily to satisfy legal requirements. Such tokenism can unintentionally cast doubt on women's qualifications, potentially hindering their career advancement in mid- and entry-level positions. Moreover, quotas may trigger backlash, as individuals affected by these laws react in opposition to preserve existing social norms (Bowles and Polanía-Reyes, 2012; Acemoglu and Jackson, 2017; Wheaton, 2022; Bian et al., 2024).

In this paper, we investigate whether adding a female to an all-male board translates into meaningful career gains for women in lower organizational positions. Understanding this relationship is crucial, as it can reveal whether gender diversity at the board level promotes broader organizational changes that support women's advancement, thus fostering a more equitable corporate environment from the ground up.

We use novel employee-employer-matched data to study various dimensions of career progression. Beyond examining women's representation within firms, our novel employer-employee-matched dataset lets us investigate shifts in hiring and promotion practices for women within treated firms. By analyzing changes in hiring and promotion patterns, we can assess whether gender quotas lead to tangible improvements in women's advancement opportunities within organizations. Moreover, our dataset enables us to explore another crucial aspect: the impact of the gender quota on women's turnover within treated firms. This helps us understand how gender quotas influence not only the advancement but also the retention of female talent within organizations. Analyzing these dynamics provides a comprehensive understanding of how quotas foster gender equality and inclusion in the workplace.

A major challenge in studying the impact of female board representation is that the decision to appoint women to boards is typically endogenous and closely tied to a company's culture toward gender diversity, which can, in turn, influence the career progression of women in junior roles. Furthermore, even when an exogenous policy change mandates board diversity, it can be difficult to identify clear control groups, as these policies often apply uniformly across companies. To overcome these challenges, we leverage the California

gender quota as a natural experiment (Hwang et al., 2018; Gertsberg et al., 2021; Greene et al., 2020; Bian et al., 2024). The quota affects only firms headquartered in a single U.S. state. This allows us to use a difference-in-differences estimator to examine changes in women's representation in treated firms.

We rely on the parallel trends assumption, which asserts that, absent the California gender quota, women's career progression in entry-level and mid-management roles would have followed similar trends over time in both treated firms (all-male board firms headquartered in California) and control firms (all-male board firms headquartered outside California). In other words, we assume that any differences in career outcomes for women post-quota implementation in treated firms can be attributed to the quota rather than to pre-existing trends.

A potential concern is that firms headquartered in California and those headquartered in other states may have had different pre-existing trends in factors that could independently affect women's career progression. This could raise doubts about whether the observed changes in treated firms are truly due to the quota or reflect pre-existing differences between California firms and those elsewhere.

To address this concern, we adopt several strategies. First, we include year fixed effects to control for time trends and ensure these changes do not confound our results. Additionally, firm fixed effects allow us to account for time-invariant characteristics unique to each firm that might influence women's career progression. We also conducted pre-trend tests to verify that treated and control firms had similar trends in female representation and career progression before the quota's implementation. This further supports the parallel trends assumption.

The California board gender quota significantly increased the number of women on corporate boards (Figure A1 and Figure 1). Our findings suggest that the quota also enhanced women's career progression in entry-level and mid-management positions.

Our main result is that treated firms saw a substantial increase in the number of female

employees, female hiring, and female promotions. Specifically, firms affected by the quota experienced a 16.7 percentage point increase in the number of female employees relative to their 2017 female workforce. Given that the average Californian firm in our sample had 53 female employees in 2017, this translates into approximately 8.9 additional female employees per firm compared to non-California firms.

Additionally, relative to non-California firms, California firms experience post-2018 increases of 10.3 percentage points in female hires, 1.9 percentage points in female promotions, and 1.7 percentage points in female exits, each scaled by the firm's 2017 female workforce. Because this denominator is fixed, the increase in exits may reflect a larger female workforce rather than a higher propensity to leave. To distinguish these explanations, we also examine female exits relative to contemporaneous female employment. The estimated effect on this exit rate is close to zero and statistically insignificant, consistent with workforce expansion contributing to the increase in exits.

Hiring increases across top-management, mid-management, and entry-level positions, while gains in promotions are concentrated in mid-management. The rise in hiring and promotions suggests that board gender quotas can expand women's employment opportunities and support career advancement beyond the boardroom. Exit rates rise among women in top management, potentially reflecting improved outside opportunities and greater labor mobility as female board representation increases. In contrast, lower exit rates among entry-level women suggest improved retention.

To investigate which types of firms respond to the gender quota, we analyze whether the quota has differential effects in male-dominated industries versus those with higher female representation. This approach allows us to determine whether such policies are effective only in sectors where women have historically been present or whether they can also drive change in industries where women have traditionally been underrepresented. We find that the effects are more salient in male-dominated industries. One possible reason is that non-male-dominated industries already have women-friendly hiring and promotion

practices, with limited scope for improvement.

We hypothesize that board-level gender quotas influence firm-level gender dynamics indirectly by shaping policies and structural changes that support women's career progression. To explore these mechanisms, we analyze employee ratings from Glassdoor across several dimensions before and after the policy implementation. Employees rate various policies and initiatives out of 5. We find improvements in compensation and benefits, work-life balance, and career opportunities. To corroborate this interpretation, we examine whether the increases in female hiring, promotion, and retention are more salient in firms with female board members serving on special committees. If so, this would suggest that the results are driven by company-wide policy changes initiated at the board level. Consistent with this mechanism, we find that the effects are particularly pronounced in firms with a female board member serving on a committee. We also find that the improvement in Glassdoor ratings is higher in firms with women on the committees. Taken together, our findings suggest that female representation on boards can improve women's career opportunities within firms, particularly when female directors have the authority to influence firm policies.

To address concerns that the increase in female employees comes from an increase in overall hiring at treated firms, we show that the number of total employees, employees hired, employees promoted, and employees exiting has no significant change or a negative change in treated firms after the quota.

We also address concerns that our treated firms are fundamentally different from those in our control in terms of their operations and attitudes towards gender equality. To mitigate these concerns, we do several things. First, we show that firms in our treated sample are similar to those in our control across various observable characteristics (Table 1). Secondly, to mitigate concerns about "attitude towards gender equality", we restrict our control sample to the "Top 20 states for working women" (excluding California), ranked by Oxfam (2022). We find that our results are robust to this specification. Finally, we show

that our findings are robust to restricting the sample to non-technology firms, which helps mitigate concerns that the results are driven by California's concentration of technology firms or by industry-specific differences in hiring, promotion, and retention practices.

Next, we address concerns about contemporaneous changes in California labor laws, including the salary history ban (AB 168), the parental leave expansion (SB 63), and pay transparency requirements (SB 1252), which may affect workers located in California regardless of firm headquarters. A key concern is that firms headquartered in California may have a larger share of their workforce located in California, and therefore a larger fraction of their employees may be directly affected by these policy changes. To address this concern, we conduct two robustness tests. First, we restrict the control sample to firms with more than 10% of their employees located in California, ensuring that control firms also have meaningful exposure to California labor laws. Second, we restrict the treated sample to California-headquartered firms with more than 20% of their workforce located outside California, reducing the likelihood that our results are driven solely by changes affecting California-based workers. Our results are robust to both restrictions.

We also examine whether the improvements in women's career outcomes extend to profit-and-loss (P&L) positions. A potential concern is that increases in hiring and promotions may be concentrated in roles that offer limited opportunities for advancement into senior leadership. To address this concern, we focus on positions whose titles indicate responsibilities in general management, operations, logistics, product management, sales, or strategic planning. We find increased female employment, hiring, and promotions in P&L roles, with gains concentrated primarily in mid-management and entry-level positions. These findings suggest that the quota's effects extend to roles involving operational and commercial responsibilities that may provide a pathway to senior leadership.

Finally, we examine whether our findings are sensitive to changes in firms' financial conditions, industry-specific shocks, and alternative approaches to statistical inference. We add financial controls and industry-by-year fixed effects to account for factors that

may independently affect women's employment and career progression. We then assess sensitivity to industry clustering, state-clustered HC3 standard errors, and a state-level wild cluster bootstrap. The positive effects on female employment, hiring, and promotions persist across these specifications, although statistical significance weakens under the wild cluster bootstrap. These checks support the robustness of our main findings.

This study contributes to the growing literature on board gender diversity. Prior work has largely examined market reactions to gender quota policies, with studies such as Hwang et al. (2018), Gertsberg et al. (2021), and Greene et al. (2020) documenting declines in stock prices following quota introductions—often attributed to higher search costs, concerns over director qualifications, or the displacement of incumbents. Similar effects were observed in Norway (Johansen and Sandnes, 2008; Ahern and Dittmar, 2012). Beyond stock price reactions, Ahern and Dittmar (2012) and Matsa and Miller (2013) find that quotas can negatively affect firm performance, leading to declines in operating profits and increases in costs. More recently, Bian et al. (2024) show that quotas can also induce gender bias in job advertisements, with firms using more masculine language to attract male candidates.

Fewer papers have examined the impact of board quotas on labor outcomes. Our study is closely related to Bertrand et al. (2018), which investigates the effects of Norway's gender quota on female representation within firms. They find that while the quota increased women's representation in senior leadership, it did not significantly reduce wage gaps among women in other top roles or increase female representation in mid-management. A key reason for the difference in our findings may be contextual: Norway's 2003 law mandated that women comprise at least 40% of corporate boards, affecting a wide range of firms and requiring substantial structural changes. In contrast, California's 2018 Senate Bill 826 required public companies headquartered in the state to have at least one female director by the end of 2019, with stricter requirements for larger boards by 2021. We focus on adding a woman to an all-male board (the extensive margin), which may generate less backlash and tokenism than policies that mandate a higher overall share of women, such as Norway's

40% quota. This setting allows us to examine the effects of initial female representation on firm policies and labor outcomes without the confounding influence of large-scale structural changes. Additionally, Norway already had relatively progressive policies supporting family leave and other benefits for women, potentially limiting the scope for newly appointed female directors to drive further policy or organizational changes that would benefit women in the broader workforce. In contrast, the United States lacks comparable legal mandates for female-friendly workplace policies, which may provide U.S. female directors with greater opportunity to implement changes that improve work environments for women.

Our study extends the existing literature in two key ways. First, we leverage novel data to examine a broad set of employment outcomes —hiring, promotions, and turnover —providing a more comprehensive view of how board quotas affect women's career trajectories. Second, we investigate heterogeneity across positions, distinguishing between mid-management and entry-level roles, which enables us to identify where the effects of board gender diversity are strongest and how they vary within the firm. Third, we provide insight into the mechanisms and channels through which the policy operates, shedding light on how female representation on boards translates into changes in organizational practices and labor outcomes.

This paper broadly adds to the literature on gender differences in labor market outcomes (Petersen and Saporta, 2004; Goldin, 2014; Bertrand et al., 2010; Blau and Kahn, 2017; Olivetti and Petrongolo, 2016). Blau and Kahn (2017) shows that barriers to women's career progression explain almost 70% of the overall gender wage gap. A key factor is the underrepresentation of women in senior positions, which widens pay differences. Similarly, Petersen and Saporta (2004) finds that in the United States, most of the wage gap comes from women and men being concentrated in different job levels, rather than from unequal pay within the same jobs. Building on this work, we examine whether increasing female representation in leadership roles improves women's promotion rates across the corporate hierarchy.

The remainder of this paper is organized as follows: Section 2 discusses the details of the gender quota; Section 3 describes our data; Section 4 describes our empirical methodology; and Section 5 summarizes our results.

2 Background

In the United States, California pioneered mandatory gender quotas when SB826 (2018) was signed into law by Governor Jerry Brown on September 30, 2018. The bill required all publicly held firms with principal executive offices in California, as reported in their SEC 10-K filings, to have at least one female director by the end of 2019. Firms with five directors were required to have two female directors, and boards with six or more directors were required to have three female directors by the end of 2021. Noncompliant firms faced fines of $100,000 for the first violation and $300,000 for subsequent violations, up to a maximum of $900,000 per year. While modest relative to the size of most publicly held firms, these penalties could create shareholder pressure to comply.

The law was overturned in May 2022 after a group of California taxpayers challenged the fines, arguing that the law violated the Equal Protection Clause and Article I, Section 31 of the California Constitution, which prohibits sex-based discrimination in public employment, education, or contracting. Although the lawsuit originally sought only to remove the fines, the lengthy court proceedings ultimately resulted in the law's reversal.

Our study focuses on firms with all-male boards that were required to add at least one female director. In these cases, the appointment often reflects a genuine shift in attitudes or openness to gender diversity, with the first female director breaking the all-male norm and gaining visibility and influence. In contrast, firms moving from one female director to multiple women were typically complying with the law, meaning that existing board members' attitudes may remain unchanged or even generate backlash. Moreover, the mandate to increase board representation beyond one woman had a longer implementation period, extending to 2021. Because the quota was challenged in court before the 2021

deadline, resistance may have been amplified. The extended timeline can reduce urgency, encourage strategic delay, and weaken both internal and external signaling of change, all of which may limit the influence of additional female directors on hiring practices and broader organizational policies. We focus on the first appointment to avoid confounding effects and to isolate the impact of genuine attitudinal shifts among board members.

SB 826, therefore, provides a novel setting for studying the effects of gender quotas, helping overcome the challenges highlighted by Ferrari et al. (2022). One such challenge is accurately identifying the timing of the policy shock. For example, in Norway, the gender quota law was passed in 2003, penalties were not established until 2005, and firms had until the end of 2007 to comply. Studies using this shock have chosen different dates for the policy effect: Ahern and Dittmar (2012) and Bertrand et al. (2018) use 2003, while Matsa and Miller (2013) uses 2006. In contrast, SB 826's relatively short and well-defined timeline—introduced in January 2018, passed in August 2018, and requiring compliance by the end of 2019—allows us to define the post-treatment period precisely.

Despite legal concerns, firms complied with the law, and the quota's impact on the gender composition of California firms was substantial. Figure 1 illustrates this effect: following the law's enactment in 2018, the proportion of board seats held by women increased significantly. It rose from 17% to 22% in 2019, further climbing to 25% by the end of 2020. By 2022, 34% of board positions were occupied by females. In contrast, non-California firms experienced a more gradual shift, moving from 17% to only 19% in 2019. However, by the end of 2022, non-California firms had reached 26% representation of women on their boards. The graph underscores a notable trend: while non-California firms increased the presence of women on their boards, the pace of change was not as rapid as observed in California firms affected by the quota. Table 2 shows that, after 2018, treated firms added approximately 0.34 more female directors and were 10 percentage points more likely to have at least one woman on the board relative to control firms. Both effects are statistically significant at the 1% level. Figure A1 illustrates how the quota impacted the distribution

of gender ratios of firms in California. Before the quota was implemented, we observed that 40% of firms had a gender ratio of 1, meaning they had only men on the board. This fell significantly after the quota was implemented.

Secondly, there was significant self-selection in Norway. Bøhren and Staubo (2014) shows that roughly 50% of the affected firms changed their organizational form. We examine whether the affected firms in California change their headquarters or delist from the stock exchange. Figure A2 & Figure A3 show the trends for headquarters changes and delistings. While Figure A4 shows the trends for IPOs. We see no increase in HQ changes, delistings, or IPOs in California after the law's passage. To further alleviate concerns regarding self-selection, we use the intent-to-treat method throughout our paper.

Similar to other papers using this shock (Hwang et al., 2018; Greene et al., 2020; Gertsberg et al., 2021; Bian et al., 2024), we leverage this natural experiment to overcome the empirical challenges of studying the impact of gender quotas.

3 Data

For our analysis, we merge data from BoardEx, Compustat, Glassdoor, and novel employer-employee-matched data to examine the relationship between director characteristics and the labor market outcomes of firm employees. Our sample consists of S&P 1500 firms.

3.1 BoardEx Data

We use BoardEx to find information about a firm's directors. BoardEx provides us with director characteristics like age, gender, nationality, and network size. It also provides us with board characteristics such as gender ratios, nationality mix, and the number of directors. We use the gender ratio reported in BoardEx to determine whether a firm had an all-male board.

3.2 Compustat and Notre Dame SRAF

We merge BoardEx data with the Notre Dame Software Repository for Accounting and Finance (SRAF) 10-X Header Data to identify the firm's headquarters location in our sample. If the HQ is missing in the SRAF, we fill it in with HQ from Compustat data. Firms with headquarters in California are included in our treated group, while firms with headquarters in other states are in our control group. We also merge with Compustat to get firm financial data to compare whether the firms in our treated and control groups have similar financial characteristics.

3.3 Employee-Employer Data

We use matched employer–employee data from Bright Data, drawn from publicly available LinkedIn profiles, to identify individuals currently or previously employed at firms with all-male boards in 2017. Using employment histories—including job titles, employment dates, and employer names—we construct a firm–year–individual panel. Due to the absence of gender information in our data, we utilize the Gendered API website, which uses a database of international names to assign gender.

We classify employees into three categories based on their job titles. Top management includes C-suite employees and Vice Presidents. We exclude CEOs and Presidents because they may serve on corporate boards. Mid-management includes Directors and Managers. Early-career employees include Analysts, Junior Software Developers, and Customer Support Representatives.

We identify a hire when an individual changes employer. We define a promotion broadly as a change in position within the same firm, encompassing both vertical advancement and lateral transitions. This measure assumes that demotions are rare. Including lateral transitions captures internal career opportunities that may involve new responsibilities and skill development without an immediate increase in hierarchical rank. Consistent with this

interpretation, Bidwell and Keller (2025) document an association between lateral mobility and subsequent promotions and pay growth.

3.4 Glassdoor Data

Glassdoor is an online platform that enables current and former employees to anonymously review their employers, offering feedback on aspects like company culture, salaries, interview processes, senior leadership, and benefits. Employees voluntarily contribute reviews, similar to how users post restaurant reviews on Yelp. To encourage contributions, Glassdoor requires new users to submit a review before accessing certain parts of the site. Glassdoor verifies reviewers via email or social media accounts to prevent self-promotion and uses rigorous fraud-detection algorithms. Reviews feature one-to-five star ratings for overall employer quality and ratings for specific categories such as Career Opportunities, Compensation and Benefits, Work-Life Balance, Senior Management, and Culture and Values.

4 Empirical Methodology

4.1 Identification Strategy

To measure the impact of gender quotas, we estimate a diff-in-diff specification. The quota was passed in 2018; hence, we define our pre-period as 2015-2018 and our post-period as 2019-2022. The quota affected firms in California with no women on the board at the time of passage. Hence, we define our treated group as firms headquartered in California with no women on the board. To ensure comparability, we designate firms headquartered outside California with all-male boards as our control group. Given variations in the timing of board statistic reporting, we define our control and treated groups based on the board gender ratio reported in 2017. To mitigate concerns about strategic firm relocation to avoid the gender quota, we employ the intent-to-treat method and rely on the headquarters

locations reported in 2017.

We estimate the following regression:

$$Y_{it} = \beta_0 + \beta_1 CaliforniaHQ_i \times Post2018_t + \alpha_i + \delta_t + \mu_{it} \tag{1}$$

Where Y_{it} measures the career progression of women for a firm i at time t. $CaliforniaHQ$ is an indicator variable for whether a firm i has headquarters in California. $Post2018$ is an indicator variable for years after 2018. We add firm fixed effects, α, to control for time-invariant differences in a firm's hiring and promoting practices. We also add year fixed effects to control for trends in hiring and promoting practices over time. We adjust for serial correlations by clustering the standard errors at the state level. We are interested in several Y variables.

We begin by examining changes in an *index of female employment* within firms, defined as

$$Y_{i\ell t} = \frac{\text{Women}_{i\ell t}}{\text{Women}_{i,2017}},$$

where $\ell \in \{\text{total}, \text{top}, \text{mid}, \text{entry}\}$. This construction fixes the denominator at each firm's 2017 female headcount, so a value of 1 corresponds to the firm's 2017 level, and estimated coefficients capture changes *relative to* that baseline (for instance, a coefficient of 0.169 indicates 16.9 additional women per 100 women employed by the firm in 2017).

We use this index rather than contemporaneous *shares* (e.g., the proportion of women among new hires) because share measures are undefined when a firm records zero hires, promotions, or exits at a given level in a given year. Using the index avoids dropping such firm-year observations and provides a consistent scale for comparing outcomes across firms and job levels. This normalization by the 2017 female workforce in a firm is applied throughout the paper in all figures and tables. All firms have positive values for the number of women employed in 2017.

We begin by examining changes in an *index of female employment* within firms. If board quotas positively impact early- and mid-career women, we expect to see a larger

increase in this index for mid-management and entry-level positions within the treated group after 2018.

Additionally, we examine trends in hiring and promoting women in firms affected by the gender quota. Hiring and promotion outcomes are constructed analogously: the number of women hired or promoted at each level is divided by the firm's total number of female employees in 2017. If the gender quota has a positive influence on these outcomes, we expect to observe larger increases in both hiring and promotion indices for the treated group after 2018.

Finally, we examine the quota's effect on women's departures using two measures. First, we scale female departures by the firm's total number of female employees in 2017, measuring exits relative to its pre-treatment female workforce. Second, we calculate the exit rate by dividing female departures in each category by contemporaneous female employment in that category. This measure helps distinguish changes in women's propensity to leave from increases in departures driven by growth in the female workforce.

We estimate each specification for top management, mid management, and entry-level positions. The key assumption underlying the causal interpretation of estimates is that, absent the quota, the trends in hiring, promotion, and turnover of women would have followed a similar trajectory in our control and treated firms (the parallel trend assumption). To test this assumption we conduct event studies to examine the trends in outcome variables before implementing the gender quota. The treated and control firms exhibit similar pre-treatment trends. This strengthens the causal interpretation of our findings by suggesting that any post-treatment divergence in outcomes is due to the intervention rather than pre-existing differences. We plot the coefficients from the event-study specification to visually inspect the parallel trends assumption. The parallel trend plots are found in the Figures section.

4.2 Descriptive Statistics

Table 1 summarizes our sample. We compared our control and treated groups on various observable characteristics before implementing the quota to mitigate concerns about potential differences between the two groups.

One concern is that our control and treated firms have different financial performance and financial constraints, which could affect their diversity initiatives. To address this concern, the first section of the table examines whether control and treated firms differ across a wide range of financial characteristics. Our analysis reveals no statistically significant disparities in financial characteristics between the two groups, providing reassurance in the comparability of their financial profiles.

Another potential concern may arise from differences between the boards of the two groups. To investigate this, we examine the board characteristics of these firms before the intervention. Our analysis indicates that the boards share several characteristics. However, they differ in two areas: the number of qualifications and board size. Firms in our treated sample tend to be slightly smaller and have higher levels of qualifications.

In the third section of the table, we test whether our control and treated samples exhibit similar distributions of observations across each position level. In the fourth section, we show that the percentage of women in each of these roles is similar for the treated and control firms before the shock. In the fifth section, we show that the number of women in each role is similar across our treated and control firms.

Finally, in the last section of the table, we show that the hiring and promotion practices, as well as the turnover rate for women, are similar for these firms before the implementation of the gender quota.

To ensure robustness, we also compared the means of the outcome variables for our treated and control groups before the quota was implemented (Table A2). Unlike Table 1, Table A2 presents the levels of these flows relative to each firm's 2017 female workforce. We find that the means are similar across most outcome variables, and the difference is not

statistically significant at 0.05 level. We find differences in overall and entry level female exits.

5 Results

5.1 The Effect of the Quota on Board Gender Composition

We first examine whether the quota changed board gender composition among firms in our employment sample. Table 2 reports estimates from regressions with firm and year fixed effects and standard errors clustered at the state level. Following the quota, California-headquartered firms experienced an increase of approximately 0.34 female directors relative to non-California firms (Column 1; $p < 0.01$). The probability of having at least one woman on the board also increased by 10.0 percentage points (Column 2; $p < 0.01$). These results establish that the quota increased female board representation in the firms used to study women's employment outcomes, providing a basis for examining whether its effects extend beyond the boardroom.

5.2 Baseline Results

5.2.1 Overall Effects on Women's Career Outcomes

Our analysis reveals that firms headquartered in California, subject to the board quota mandate, exhibit significant improvements in both the recruitment and promotion of women. The estimation results, presented in Table 3, Panel A, highlight these effects. We find significant increases in total female employment (Column 1; $p < 0.05$), female hiring (Column 2; $p < 0.01$), and female promotions (Column 3; $p < 0.05$). Specifically, firms affected by the quota experienced a 16.7 percentage-point increase in the number of female employees relative to their 2017 female workforce, a 10.3 percentage-point increase in female hires, and a 1.9 percentage-point rise in the promotions awarded to women, all

measured relative to the firm's 2017 female employee base. Additionally, we observe a significant 1.7 percentage-point increase in female exits in treated firms (Column 4; $p < 0.01$). Because exits are measured relative to the fixed 2017 female workforce, this increase may reflect growth in female employment rather than a greater propensity to leave. We therefore also examine exit rates, which scale female departures by contemporaneous female employment, and find an effect close to zero and statistically insignificant (Column 5).

For a California firm with 53 female employees in 2017, a 10.3 percentage-point increase in female hires relative to this base translates into approximately 5.5 additional female hires compared to non-California firms. Similarly, a 1.9 percentage-point rise in female promotions corresponds to approximately 1 additional promotion, while a 1.7 percentage-point increase in female exits equates to approximately 0.9 additional female departures.

Together, these results indicate that the gender quota increases women's representation and career progression within firms. These relative increases coincide with the implementation of the gender quota (Figure 2). The event-study patterns provide support for the parallel trends assumption for total female employment, hiring, and promotions, although some pre-trends are evident in exits.

5.2.2 Effects Across Organizational Levels

Next, we examine whether these effects differ across top-management, mid-management, and entry-level positions.

We first examine the impact of the board quota on women in top management roles. We use our employer–employee matched data for this analysis and present our findings in Table 3, Panel B. Our findings indicate a 5.6 percentage-point increase in the total number of women in top management (Column 1; $p < 0.01$) and a 1.2 percentage-point rise in female hires at this level (Column 2; $p < 0.01$). The estimated effect on female promotions is negative but close to zero and is statistically insignificant (Column 3). We also observe a 0.3 percentage-point increase in exits among women in top management

(Column 4; $p < 0.05$). These outcomes are measured relative to the firm's 2017 female workforce. For a California firm with 53 female employees in 2017, these changes correspond to approximately 3 additional women in top management, 0.6 additional female hires, and 0.2 additional female departures. We also find a 0.4 percentage-point increase in the exit rate among women in top management (Column 5; $p < 0.01$), suggesting that departures at this level rise beyond what would be explained by workforce expansion alone.

We also assess the impact of the board quota on women in mid-management roles, with results presented in Table 3, Panel C. Our analysis reveals a significant 5.6 percentage-point increase in the number of women in mid-management positions (Column 1; $p < 0.05$). Additionally, we find a notable 3.4 percentage-point rise in the number of female hires (Column 2; $p < 0.01$) and a 1.0 percentage-point increase in female promotions (Column 3; $p < 0.01$). These effects are economically meaningful. For a California firm with 53 female employees in 2017, a 5.6 percentage-point increase relative to this base corresponds to approximately 3 additional female mid-managers. Similarly, the 3.4 percentage-point rise in female hiring translates into approximately 1.8 additional women hired into mid-management roles relative to non-California firms. The 1.0 percentage-point increase in female promotions corresponds to approximately 0.5 additional women promoted. We also estimate a 0.2 percentage-point increase in female exits (Column 4) and a 0.1 percentage-point increase in the exit rate (Column 5), although neither estimate is statistically significant. These relative increases coincide with the implementation of the gender quota, as shown in Figure 3.

Finally, we study the impact of the board quota on women early in their careers (Table 3, Panel D). We observe significant changes in female hires (Column 2; $p < 0.05$) and exits (Column 4; $p < 0.01$). The estimated increases in total female entry-level employment and promotions are 5.3 and 0.4 percentage points, respectively, relative to the firm's 2017 female workforce, but neither is statistically significant (Columns 1 and 3). We find a 5.7 percentage-point increase in female hires (Column 2; $p < 0.05$), corresponding to

approximately 3 additional women hired into early-career roles for a firm with 53 female employees in 2017. We also observe a 1.1 percentage-point increase in female exits relative to the 2017 female workforce (Column 4; $p < 0.01$), translating into approximately 0.6 additional female departures. However, the exit rate declines by 0.5 percentage points (Column 5; $p < 0.05$), suggesting improved retention despite the increase in departures relative to the fixed pre-policy workforce. To assess the parallel trends assumption, we examine whether outcomes evolve similarly for treated and control firms before the introduction of the 2018 policy. Figure 4 provides support for parallel trends in hiring, although some pre-trends are evident in total female exits.

Together, these results suggest that the quota improves women's career outcomes across organizational levels. Hiring increases in top-management, mid-management, and entry-level positions, while promotion gains are concentrated in mid-management. The differing exit-rate patterns suggest that these gains may operate through different channels across the corporate hierarchy. Higher exit rates among women in top management may reflect improved outside opportunities and greater labor mobility. In contrast, lower exit rates among entry-level women are consistent with improved retention, potentially reflecting better workplace conditions or stronger prospects for internal advancement.

5.3 Heterogeneity by Industry Gender Composition

Next, we investigate how the quota impacts firms in male-dominated industries compared to those in non-male-dominated industries. We define "male-dominated" industries as those with 60% or more male workers in 2017, according to the Bureau of Labor Statistics. On the one hand, we may expect weaker results in male-dominant industries, as there may not be enough women in the industry's labor force. On the other hand, we may expect stronger results in these industries, as non-male-dominant industries might already have women-friendly policies and require little reform. We first formally examine whether the quota's effects differ between male-dominated and non-male-dominated industries using a

triple-interaction specification (Table 4). The coefficient on $CaliforniaHQ \times Post2018$ captures the effect in non-male-dominated industries. The triple-interaction coefficient, $CaliforniaHQ \times Post2018 \times MD$, where MD indicates male-dominated industries, captures the difference in this effect between male-dominated and non-male-dominated industries. Adding these two coefficients gives the total effect for firms in male-dominated industries.

Panel A reports the overall employment outcomes. The triple-interaction estimates indicate a 14.7 percentage-point smaller effect on female employment, a 0.5 percentage-point larger effect on hiring, and a 0.3 percentage-point larger effect on promotions in male-dominated industries, although none of these differences is statistically significant (Columns 1–3). However, the effect on female exits relative to the 2017 female workforce is 8.1 percentage points lower in male-dominated industries (Column 4; $p < 0.01$). The effect on the female exit rate is also 3.0 percentage points lower (Column 5; $p < 0.01$). The total effects for male-dominated firms indicate increases of 14.5 percentage points in female employment (Column 1; $p < 0.10$), 10.7 percentage points in hiring (Column 2; $p < 0.01$), and 2.1 percentage points in promotions (Column 3; $p < 0.05$). Female exits decline by an estimated 0.2 percentage points, although this effect is statistically insignificant (Column 4), while the exit rate decreases by 0.7 percentage points (Column 5; $p < 0.05$). Thus, although the differences in employment, hiring, and promotion effects are statistically insignificant, the retention effects differ significantly across industry groups.

We next examine how these differences vary across organizational levels. Among women in top management (Panel B), the quota's effect on total employment is 4.3 percentage points lower in male-dominated industries (Column 1; $p < 0.01$). However, its effects on hiring and promotions are 2.4 and 1.2 percentage points higher, respectively (Columns 2–3; $p < 0.01$). The effect on exits is 0.3 percentage points lower but statistically insignificant (Column 4), while the effect on the exit rate is 0.4 percentage points lower (Column 5; $p < 0.05$). The total effects for male-dominated firms indicate increases of 3.8 percentage points

in female top-management employment, 1.6 percentage points in hiring, and 0.2 percentage points in promotions (Columns 1–3; $p < 0.01$). The total effects on exits and exit rates are both positive, at 0.3 percentage points, but statistically insignificant (Columns 4–5). The negative triple-interaction coefficient for employment therefore indicates a smaller increase in male-dominated industries, rather than a decline in female top-management employment.

Among women in mid-management (Panel C), the effects on employment, hiring, and promotions are larger in male-dominated industries. The triple-interaction estimates indicate differences of 12.2 percentage points in female employment (Column 1; $p < 0.05$), 8.7 percentage points in hiring (Column 2; $p < 0.01$), and 1.8 percentage points in promotions (Column 3; $p < 0.01$). The differences in exits and exit rates are an estimated decrease of 0.4 percentage points and an increase of 0.1 percentage points, respectively, neither of which is statistically significant (Columns 4–5). The total effects for male-dominated firms indicate increases of 9.3 percentage points in female employment, 5.5 percentage points in hiring, and 1.5 percentage points in promotions (Columns 1–3; $p < 0.01$). The total effects on exits and exit rates are increases of 0.2 and 0.1 percentage points, respectively, but neither is statistically significant (Columns 4–5).

For entry-level women (Panel D), the effects on employment and hiring are smaller in male-dominated industries by 21.6 and 10.4 percentage points, respectively (Columns 1–2; $p < 0.05$). The effect on promotions is also 1.4 percentage points lower, although this difference is statistically insignificant (Column 3). The effects on exits and exit rates are lower by 7.4 and 2.7 percentage points, respectively (Columns 4–5; $p < 0.01$). The total effects for male-dominated firms indicate increases of 1.5 percentage points in entry-level employment, 3.6 percentage points in hiring, and 0.1 percentage points in promotions, none of which is statistically significant (Columns 1–3). In contrast, exits relative to the 2017 female workforce decline by 0.6 percentage points (Column 4; $p < 0.10$), and the exit rate decreases by 1.1 percentage points (Column 5; $p < 0.01$). These results suggest that

the benefits for entry-level women in male-dominated industries are particularly evident in improved retention.

We next examine the industry subsamples separately. Table A3 reports results for male-dominated industries. Panel A shows increases in female employment, hiring, and promotions, alongside a decline in the overall female exit rate. Across organizational levels, the increases in employment, hiring, and promotions are evident in both top management (Panel B) and mid-management (Panel C). Among entry-level women (Panel D), employment, hiring, and promotion estimates are positive but statistically insignificant, while exits and exit rates decline. These subsample results reinforce the pattern of managerial career gains and improved entry-level retention.

Table A4 reports results for non-male-dominated industries. Panel A shows a 29.2 percentage-point increase in female employment (Column 1; $p < 0.05$) and a 10.2 percentage-point increase in hiring (Column 2; $p < 0.10$). Promotions increase by an estimated 1.8 percentage points, but this effect is statistically insignificant (Column 3). Female exits and exit rates increase by 7.9 and 2.3 percentage points, respectively (Columns 4–5; $p < 0.01$). Among women in top management (Panel B), employment increases by 8.1 percentage points (Column 1; $p < 0.01$), while hiring declines by an insignificant 0.8 percentage points (Column 2) and promotions decline by 1.0 percentage point (Column 3; $p < 0.01$). Exits and exit rates increase by 0.6 and 0.7 percentage points, respectively (Columns 4–5; $p < 0.01$).

Among mid-managers in non-male-dominated industries (Panel C), employment and promotions decline by an estimated 2.9 and 0.2 percentage points, respectively, but neither effect is statistically significant (Columns 1 and 3). Hiring decreases by 3.2 percentage points (Column 2; $p < 0.05$), while exits increase by 0.5 percentage points (Column 4; $p < 0.01$). The estimated 0.1 percentage-point increase in the exit rate is statistically insignificant (Column 5). Among entry-level women (Panel D), employment increases by 23.1 percentage points (Column 1; $p < 0.05$) and hiring increases by 14.0 percentage

points (Column 2; $p < 0.01$). Promotions increase by an insignificant 1.5 percentage points (Column 3). Exits rise by 6.8 percentage points (Column 4; $p < 0.01$), and the exit rate increases by 1.5 percentage points (Column 5; $p < 0.05$).

Taken together, the results suggest that the quota's overall effects on female employment, hiring, and promotions do not differ significantly between male-dominated and non-male-dominated industries. However, the effects on retention and on specific career stages differ. Non-male-dominated industries experience larger increases in female employment in top-management and entry-level positions, as well as larger increases in entry-level hiring. Male-dominated industries experience larger gains in mid-management employment, hiring, and promotions, alongside more positive hiring and promotion effects in top management. Entry-level retention also improves in male-dominated industries, whereas exit rates increase in non-male-dominated industries. These patterns suggest that industry composition shapes where women experience gains in employment, career advancement, and retention following board gender quotas.

5.4 Mechanisms

The baseline results show that the quota increased women's hiring and promotions, this is particularly evident in managerial positions. In this section, we examine potential mechanisms through which the policy may translate into workforce outcomes. We consider two related channels. First, we examine whether treated firms experience changes in workplace practices as perceived by their employees following the quota. Second, we examine whether the effects on women's career outcomes are stronger when female directors serve on board committees, where they may have greater scope to influence firm policies and organizational decisions.

5.4.1 Changes in Workplace Practices and Employee Experience

To examine whether the quota was accompanied by broader changes in the workplace environment, we analyze employee ratings from Glassdoor across several dimensions before and after the policy's implementation (Table 5). Our analysis reveals significant improvements in overall ratings, culture and values, compensation and benefits, career opportunities, and senior leadership. These changes suggest an enhanced work environment supporting women's career advancement and retention.

We observe a 0.24-star increase in Culture and Values (column 2, $p< 0.01$), which suggests that the gender quota may have influenced a broader cultural shift within these companies, fostering an environment that values diversity and inclusion. A supportive culture can make organizations more attractive to female talent and encourage them to remain within the company, aligning with the increased retention rates we observe for women.

We also observe a 0.09 star increase in Compensation and Benefits (column 3, $p< 0.01$), which could imply more equitable compensation practices, potentially addressing gender pay disparities and making these workplaces more appealing for women. This enhanced equity could drive both recruitment and retention of female employees.

We see a 0.17 star increase in the Career Opportunities (column 4, $p< 0.05$) rating, which suggests potentially more career advancement and skill development pathways within the firm. For women, this improvement can be particularly impactful as it may reduce barriers to upward mobility by providing clearer pathways for promotions, access to mentorship programs, and more robust professional development resources.

We see a 0.18 star increase in work-life balance (column 5, $p< 0.01$) rating, suggesting that firms may have introduced more flexible work arrangements, supportive policies, or better work-life integration. For women, this improvement can be particularly impactful, as greater flexibility and a healthier work-life balance are often critical for career advancement.

Finally, we also see a 0.28 star increase in the ratings for Senior Leadership (column 6,

p< 0.01). This suggests that employees may be responding favorably to the management style and contributions of female leaders, which could contribute to reducing biases against women across the organization.

5.4.2 Female Directors' Participation in Board Committees

We next examine whether the effects on women's career outcomes are stronger when female directors hold positions that may give them greater influence over firm policies and organizational decisions. Table 6, Panel A examines whether the effects of the California board quota are stronger in firms where female board members serve on committees. The interaction term of interest, *CaliforniaHQ* × *Post2018* × *Committee2019*, indicates larger effects in these firms: 40.4 percentage points for female employment (Column 1; $p < 0.10$), 16.6 percentage points for female hiring (Column 2; $p < 0.05$), and 8.5 percentage points for female promotions (Column 3; $p < 0.01$). The effects on female exits and exit rates are also higher by 2.6 and 1.1 percentage points, respectively (Columns 4–5; $p < 0.05$). In addition, the effect on firms' overall Glassdoor rating is approximately 0.15 stars larger in firms where women serve on board committees (Column 6; $p < 0.10$).

Adding the baseline quota effect to the triple-interaction coefficient gives the total effect for firms with women on committees. These estimates indicate increases of approximately 36.9 percentage points in female employment (Column 1), 17.4 percentage points in hiring (Column 2), 5.3 percentage points in promotions (Column 3), and 2.7 percentage points in exits (Column 4), each measured relative to the firm's 2017 female workforce. The total effect on the exit rate is an increase of approximately 0.3 percentage points (Column 5). The total effect on the overall Glassdoor rating is approximately 0.32 stars, combining the baseline effect of 0.175 with the triple-interaction coefficient of 0.148 (Column 6).

These findings indicate that the estimated effects on women's employment, hiring, and promotions are larger in firms where female board members hold committee positions. The Glassdoor results also indicate larger improvements in employees' overall perceptions of the

workplace. These patterns are consistent with the idea that committee participation gives female directors greater scope to influence company-wide policies, although they do not establish committee participation itself as the cause of these differences.

Table 6 examines whether these patterns differ across organizational levels. Among women in top management (Panel B), the triple-interaction estimates indicate a 3.5 percentage-point larger effect on hiring (Column 2; $p < 0.01$), a 0.3 percentage-point lower effect on exits (Column 4; $p < 0.01$), and a 0.1 percentage-point lower effect on the exit rate (Column 5; $p < 0.05$). The estimated differences in total female employment and promotions are positive, at 1.5 and 0.3 percentage points, respectively, but statistically insignificant (Columns 1 and 3). Adding the baseline quota effect to the triple interaction gives total increases of approximately 7.4 percentage points in female top-management employment (Column 1), 2.6 percentage points in hiring (Column 2), and 0.1 percentage points each in promotions, exits, and exit rates (Columns 3–5) for firms with women on committees.

Among women in mid-management (Panel C), the triple interaction reveals larger effects on employment and career advancement. The effect on female employment is 19.4 percentage points larger (Column 1; $p < 0.05$), while the effects on hiring and promotions are 5.2 and 4.1 percentage points larger, respectively (Columns 2–3; $p < 0.01$). The estimated differences in exits and exit rates are positive, at 0.9 and 0.3 percentage points, respectively, but neither is statistically significant (Columns 4–5). The total effects for firms with women on committees indicate increases of approximately 13.8 percentage points in female mid-management employment (Column 1), 5.5 percentage points in hiring (Column 2), 2.8 percentage points in promotions (Column 3), 0.5 percentage points in exits (Column 4), and 0.3 percentage points in the exit rate (Column 5).

Finally, among entry-level women (Panel D), the triple interaction indicates a 22.5 percentage-point larger effect on female employment (Column 1; $p < 0.10$). The estimated effects on hiring and promotions are also larger, by 8.4 and 1.0 percentage points,

respectively, but neither difference is statistically significant (Columns 2–3). The effects on exits and exit rates are larger by 2.0 and 1.0 percentage points, respectively (Column 4; $p < 0.01$; Column 5; $p < 0.05$). The total effects for firms with women on committees indicate increases of approximately 17.6 percentage points in female employment (Column 1), 9.7 percentage points in hiring (Column 2), 1.0 percentage point in promotions (Column 3), and 2.0 percentage points in exits (Column 4). The total effect on the exit rate is approximately zero, as the positive triple interaction offsets the negative baseline quota effect (Column 5).

These findings are consistent with committee participation supporting women's career opportunities, particularly through stronger hiring in top management and greater employment, hiring, and promotions in mid-management. The retention patterns are more mixed. Committee participation is associated with a smaller increase in top-management exit rates, although the estimated total effect remains positive. Among entry-level women, committee participation offsets the decline in exit rates observed among firms without women on committees, leaving an estimated total effect close to zero.

5.5 Robustness

We next conduct a series of robustness exercises to assess if our baseline results hold under alternative sample restrictions, outcome definitions, or inference methods. First, we examine whether the results reflect broader changes in firm employment rather than changes in women's labor outcomes specifically. Second, we address concerns that the estimates may be driven by differences between California and the control states, including broader women-friendly policies, contemporaneous California labor laws, and California's concentration of technology firms. We also examine whether the effects occur in profit-and-loss functions, which are particularly relevant for advancement into senior leadership. Finally, we assess the sensitivity of the results to other firm controls including financial controls, industry-specific shocks, and alternative approaches to statistical inference. Across these

exercises, the main patterns of increased female employment, hiring, and career advancement remain broadly robust.

5.5.1 Robustness - Head Count

Table A5 examines whether SB 826 is associated with changes in overall workforce headcount and mobility. This test helps assess whether the observed effects on women's career progression reflect broad changes in firm employment rather than changes in the composition or allocation of workers across roles.

The estimates provide little evidence that treated firms experienced differential changes in overall employment after 2018. The coefficient on ($CaliforniaHQ \times Post2018$) is negative for total employees, but it is statistically insignificant. Similarly, the coefficients for promotions and exits are statistically insignificant. The only statistically significant effect is for total hires: California-headquartered firms reduce hiring by approximately 7.9 employees relative to control firms after 2018. This decline suggests that the main results are unlikely to be driven by a general expansion in hiring among treated firms.

Overall, the results indicate that the quota is not associated with large changes in aggregate workforce size, promotions, or exits. Instead, the evidence is more consistent with changes in the composition of career outcomes for women across organizational levels, rather than broad changes in firm-level employment dynamics.

5.5.2 Robustness - Women Friendly States

A potential concern with our findings is that the observed effects of the quota may be driven by California's broader women-friendly policies, rather than the quota itself. To address this, we restrict our control sample to the top 20 women-friendly states identified by Oxfam (2022), excluding California. Our results remain robust under this specification, reinforcing the independent impact of the board quota.

Our analysis reveals significant improvements within treated firms (Table 7, Panel A).

We find an 18.7 percentage point increase in the total number of female employees (Column 1; $p < 0.05$), a 12.9 percentage point rise in female hires (Column 2; $p < 0.01$), and a 2.1 percentage point increase in female exits (Column 4; $p < 0.01$).

For a California firm with an average of 53 female employees, these changes translate into meaningful workforce shifts. The 18.7 percentage point rise in female representation corresponds to approximately 10 additional female employees per firm, while the 12.9 percentage point increase in hiring equates to 6.8 more female hires. At the same time, the 2.1 percentage point increase in female exits reflects 1.1 additional female employees leaving the firm.

Examining top management positions (Table A6, Panel A), we find a 4.4 percentage point increase in the total number of women in executive roles (Column 1; $p < 0.01$) and a 0.8 percentage point rise in female hires (Column 2; $p < 0.05$). For a California firm with an average of 53 female employees, these changes correspond to 2.3 additional female top managers and 0.42 more female hires at the executive level.

For mid-management positions (Table A6, Panel B), we observe similar positive trends in treated firms. Our analysis shows a 3.2 percentage point increase in female hiring (Column 2; $p < 0.05$) and a 0.6 percentage point increase in female exits (Column 4; $p < 0.05$). These findings suggest that the board quota not only influenced top leadership but also contributed to increasing female representation at mid-management levels.

For entry-level positions (Table A6, Panel C), we observe similar positive trends in treated firms. Our analysis shows an 11.2 percentage point increase in the total number of women (Column 1; $p < 0.05$), an 8.6 percentage point increase in female hiring (Column 2; $p < 0.01$), and a 1.1 percentage point increase in female exits (Column 4; $p < 0.01$). These findings suggest that the board quota not only influenced top leadership but also contributed to increasing female representation at entry-level positions.

5.5.3 Robustness - Profit and Loss Functions

We examine whether the increase in women's career progression extends to profit and loss (P&L) functions, which are often viewed as central to advancement into senior leadership. We classify positions as P&L roles if their titles contain terms indicating general management, operations, logistics, product management, sales, or strategic planning.

We first examine P&L outcomes scaled by the firm's total number of female employees in 2017, consistent with our baseline analysis. Table 7, Panel B reports aggregate effects across all P&L-related positions. Relative to control firms, California-headquartered firms experience a 2.8 percentage-point increase in women in P&L positions (Column 1; $p < 0.01$), a 1.1 percentage-point increase in female hires into these roles (Column 2; $p < 0.01$), and a 0.6 percentage-point increase in female promotions (Column 3; $p < 0.01$). We find no statistically significant effects on female exits relative to the 2017 female workforce or on P&L departures relative to contemporaneous total female employment (Columns 4–5).

Table A7 decomposes these effects by organizational level. Panel B shows a 1.8 percentage-point increase in women in P&L-related mid-management positions (Column 1; $p < 0.05$) and a 0.5 percentage-point increase in hiring (Column 2; $p < 0.01$). Panel C shows a 0.8 percentage-point increase in women in entry-level P&L positions and a 0.6 percentage-point increase in hiring into these roles (Columns 1–2; $p < 0.01$). These outcomes are measured relative to the firm's total female workforce in 2017. Promotions and departures do not change significantly within these categories. Panel A shows no statistically significant effects on women in top management P&L roles.

As an additional analysis, Table A8 scales employment, hiring, promotions, and exits by the firm's number of female P&L employees in 2017. This alternative normalization measures changes relative to the firm's pre-policy female workforce in P&L functions. Exit rates use contemporaneous female P&L employment, within the relevant career category for the panel-level analysis.

Table A8, Panel A reports aggregate effects under this alternative normalization. Rel-

ative to control firms, California-headquartered firms experience a 37.1 percentage-point increase in women in P&L positions, a 7.6 percentage-point increase in female hires, and an 8.7 percentage-point increase in female promotions, all measured relative to the firm's 2017 female P&L workforce (Columns 1–3; $p < 0.01$). Although the coefficient on exits relative to the 2017 female P&L workforce is small and statistically insignificant, the exit rate increases by 0.8 percentage points (Column 5; $p < 0.01$). Thus, the gains in employment and advancement coexist with an increase in departures relative to contemporaneous female P&L employment.

Table A8 decomposes these effects by organizational level. The increase in women in P&L roles is concentrated primarily in mid-management positions. Panel C shows a 28.9 percentage-point increase in women in P&L-related mid-management roles, an insignificant 1.8 percentage-point increase in hires (Column 2), and a 3.4 percentage-point increase in promotions (Columns 1 and 3; $p < 0.01$). Panel D shows increases of 6.2 percentage points in female entry-level P&L employment (Column 1; $p < 0.05$), 6.1 percentage points in hiring (Column 2; $p < 0.01$), and 2.7 percentage points in promotions (Column 3; $p < 0.01$). In contrast, Panel B shows a smaller, 2.0 percentage-point increase in female top-management P&L employment (Column 1; $p < 0.05$), no significant increase in hiring, and a decline in promotions relative to the 2017 female P&L workforce. The top-management exit rate nevertheless increases by 0.1 percentage points (Column 5; $p < 0.01$).

Overall, both normalizations indicate that the quota is associated with increased female employment and hiring in P&L functions, particularly in mid-management and entry-level positions. The evidence for promotions within these career levels is stronger when outcomes are scaled by the pre-policy female P&L workforce. These findings are consistent with improvements in women's access to roles that may support subsequent advancement into senior leadership, although the increase in P&L exit rates indicates that these gains do not necessarily coincide with improved retention.

5.5.4 Robustness - Restricting Controls to those with more than 10% of workers in CA

Table 7, Panel C examines whether the main results are robust to restricting the control group to firms with substantial exposure to California-based workers. This test addresses the concern that contemporaneous California labor laws may affect workers located in California regardless of firm headquarters. If California-headquartered firms have a larger share of employees located in California, then changes in women's career outcomes could partly reflect exposure to other California labor policies rather than the board gender quota itself. To mitigate this concern, we restrict the control group to firms with 10% or more employees in California.

The results remain broadly consistent with the baseline findings. In Table 7, Panel C, the coefficient on $(CaliforniaHQ \times Post2018)$ is positive across all four outcomes. Treated firms experience a statistically significant increase in total female employees, female hires, and female exits, relative to the restricted control group. The hiring effect is especially strong: California-headquartered firms increase female hires by 15 percentage points relative to their 2017 female employment base. The coefficient on female promotions is also positive, although it is not statistically significant. These estimates suggest that the main findings are not driven solely by comparing California-headquartered firms to control firms with little exposure to California labor markets.

Table A9 further decomposes the effects by organizational level. Panel A shows that the increase in women in top management remains positive and statistically significant. Treated firms experience increases in total women in top management and top-management hires, while the effect on top-management promotions is small and statistically insignificant. Panel B shows positive effects for mid-management hiring and exits, although the effect on total mid-management representation is not statistically significant. Panel C shows that the largest effects appear among early-career employees, where treated firms experience significant increases in female hires and exits, and a positive increase in total early-career

female employment.

Overall, the results indicate that the main findings are robust to using a more geographically comparable control group. The effects are not explained solely by control firms having limited exposure to California-based workers. Instead, the evidence continues to suggest that California-headquartered firms experienced an increase in women's career progression after the implementation of SB 826, with particularly robust effects on hiring and representation in top management and early-career roles.

5.5.5 Robustness - Restrict Treated firms to those with 20% or more workers outside CA

Tables 7, Panel D and A10 further address concerns that the results may be driven by contemporaneous California labor laws that affect employees located in California, rather than by the board gender quota itself. Since California-headquartered firms may have a larger share of their workforce located in California, other state-level labor policy changes could differentially affect treated firms. To mitigate this concern, we restrict the treated sample to California-headquartered firms with at least 20% of their workforce located outside California in 2017. This restriction reduces the likelihood that the results are driven only by changes affecting California-based workers.

Table 7, Panel D shows that the results remain positive and statistically significant in this restricted treated sample. The coefficient on $(CaliforniaHQ \times Post2018)$ is positive and significant for total female employment, female hiring, and female promotions. Treated firms experience a 0.163 increase in female employees relative to their 2017 female employment base, along with a 0.059 increase in female hires and a 0.033 increase in female promotions. The coefficient on female exits is negative and statistically insignificant, and the exit rate declines significantly, suggesting that the increase in women's representation is not driven by greater turnover.

Table A10 decomposes these effects by organizational level. Panel A shows that the

effect remains positive and statistically significant for women in top management. Treated firms experience increases in total women in top management, top-management hires, and top-management promotions. Panel B shows similar evidence for mid-management, with significant increases in total women in mid-management, mid-management hires, and mid-management promotions. Panel C shows weaker effects for early-career roles: the coefficients on total early-career women and hires are positive but not statistically significant, while promotions increase significantly and exits and the exit rate decline significantly.

Overall, these results suggest that the main findings are not driven solely by California-headquartered firms having a larger share of workers located within California. Even when treated firms with highly California-concentrated workforces are excluded, the estimates remain positive and significant for overall female employment, hiring, and promotions. The decomposition further suggests that the effects are especially concentrated in top management and mid-management roles, consistent with the quota improving women's advancement within the managerial pipeline rather than simply reflecting broad changes in California-based employment.

5.5.6 Robustness - Excluding Tech Firms

Tables 7, Panel E and A11 examine whether the main results are robust to excluding technology firms. We define technology firms as those classified by BoardEx in the "Software and Computer Services," "Information Technology," or "Hardware" sectors. This test addresses the concern that the estimated effects may be driven by California's concentration of technology firms, which may differ from other firms in their workforce structure, hiring practices, growth trajectories, or baseline attitudes toward gender diversity. By restricting the sample to non-tech firms, we test whether the relationship between SB 826 and women's career progression persists outside the technology sector.

Table 7, Panel E shows that the results remain strong in the non-tech firm sample. The coefficient on $(CaliforniaHQ \times Post2018)$ is positive and statistically significant

across all four outcomes. Relative to control firms, California-headquartered non-tech firms experience a 0.302 increase in total female employees, scaled by the number of female employees in 2017. The estimates also indicate significant increases in female hiring, female promotions, and female exits. The positive effects on hiring and promotions suggest that the increase in women's representation is not limited to changes in retention or turnover, but reflects increased movement of women into and through firms after the policy.

Table A11 decomposes these effects by organizational level. Panel A shows that women's representation in top management increases significantly in non-tech firms, with positive effects on total top-management representation, hiring into top-management roles, and exits. Panel B shows positive but somewhat weaker effects for mid-management: the coefficients on total mid-management representation and promotions are positive and statistically significant at the 5 percent level, and the hiring effect is also positive and statistically significant (Column 2; $p < 0.05$), while the exit effect is not statistically significant. Panel C shows strong effects among entry-level workers, with statistically significant increases in total representation, hiring, promotions, and exits.

Overall, these results indicate that the main findings are not driven solely by technology firms. Even after excluding tech firms, California-headquartered firms experience increases in women's representation and career progression after 2018. The effects appear across multiple organizational levels, with especially strong evidence for entry-level roles and continued evidence of gains in top management. This supports the interpretation that the quota affected women's career outcomes more broadly, rather than reflecting industry-specific dynamics in the technology sector.

5.5.7 Other Robustness Tests

We examine the robustness of our results for women's employment outcomes across the firm to financial controls, industry-specific shocks, and alternative approaches to statistical inference. Table 7, Panels F–J report these specifications for total female employment,

hiring, promotions, exits, and exit rates.

First, a potential concern is that changes in firms' financial conditions influence both their demand for employees and women's career outcomes. Panel F adds controls for cash holdings, tangibility, Tobin's Q, book leverage, operating profitability, and capital expenditures. The specification also includes firm and year fixed effects. Female employment, hiring, and promotions increase by 33.7, 17.2, and 5.1 percentage points, respectively, relative to the firm's 2017 female workforce, with all three estimates significant at the 1% level. Female exits increase by 1.5 percentage points ($p < 0.05$), while the exit rate decreases by approximately 0.6 percentage points ($p < 0.10$).

Second, California's industry composition may expose its firms to sector-specific changes in employment opportunities. Panel G adds two-digit SIC industry-by-year fixed effects, absorbing annual shocks shared by firms within the same industry. The estimated effects remain positive for female employment, hiring, and promotions, at 39.1, 20.8, and 5.4 percentage points, respectively ($p < 0.01$). The effects on exits and exit rates are statistically insignificant. These results suggest that the gains in employment and career advancement are not explained solely by common industry-specific annual shocks.

Third, we examine whether statistical significance depends on how we account for correlated regression errors. Panel H clusters standard errors by two-digit SIC industry, allowing errors to be correlated across firms and years within an industry. Female employment remains significant at the 5% level, while hiring and promotions remain significant at the 1% level. Industry clustering provides an alternative assessment of within-industry dependence, although it does not directly account for shocks shared across industries within a state.

Panel I uses state-clustered HC3 standard errors without the Hansen adjustment. This approach adjusts for leverage that can affect conventional cluster-robust standard errors. The employment, hiring, and promotion estimates remain significant at the 1% level. Panel J uses a null-imposed wild cluster bootstrap with state-level weights and 10,000 repetitions

to assess sensitivity to an alternative reference distribution for inference. Hiring remains significant at the 5% level, while employment and promotions are significant at the 10% level.

Overall, the positive effects on female employment, hiring, and promotions persist across these specifications, although statistical significance weakens under the wild cluster bootstrap. The evidence for changes in exits and exit rates is less robust.

6 Conclusion

In 2018, California enacted SB 826, requiring publicly held companies headquartered in the state to include women on their boards. We examine whether the effects of this mandate extended beyond the boardroom to women's careers within firms. Using a novel employer–employee dataset, we find that the quota increased female employment, hiring, and promotions. Relative to firms' 2017 female workforce, these outcomes increased by 16.7, 10.3, and 1.9 percentage points, respectively. Although female departures also increased relative to this fixed workforce base, the overall exit-rate effect was close to zero and statistically insignificant. This distinction suggests that higher departures may partly reflect an expanding female workforce rather than a greater propensity to leave.

Furthermore, the effects varied across organizational levels. Hiring increased in top-management, mid-management, and entry-level positions, while promotion gains were concentrated in mid-management. These findings suggest that board gender quotas can influence both women's access to firms and their subsequent advancement. Retention effects were more mixed: exit rates increased among women in top management but declined among entry-level women. Higher top-management exit rates could reflect greater outside opportunities and labor mobility, although displacement or dissatisfaction could also contribute. Lower entry-level exit rates are consistent with improved retention. Thus, a policy that changes representation at the top may affect women at different career stages through different channels.

Our results also provide insight into the organizational conditions associated with these effects. The gains in women's employment, hiring, and promotions were larger when female directors served on board committees, consistent with committee participation providing greater scope to influence firm policies. Improvements in Glassdoor ratings suggest that the quota was also accompanied by changes in employees' perceptions of the workplace. Industry composition mattered as well: male-dominated industries experienced larger gains in mid-management employment and advancement, alongside more favorable entry-level retention effects. Together, these patterns suggest that female board representation may translate into different employment outcomes depending on directors' roles and the organizational environment.

Whether these gains translate into sustained improvements in women's careers remains an open question. Increased hiring and internal mobility need not lead to lasting gains in pay, responsibility, or access to senior leadership. Similarly, departures may represent either career advancement outside the firm or the loss of opportunities within it. Following employees across firms and over longer periods would help distinguish these possibilities.

Nevertheless, our findings suggest that board gender quotas can have consequences beyond the composition of corporate boards. The increases in hiring and promotions, together with gains in women's participation in profit-and-loss roles, are consistent with changes in career opportunities within firms. These benefits coexist with differences in retention across organizational levels, highlighting the importance of examining a broad set of employment outcomes when assessing board diversity policies. Whether these changes ultimately narrow gender gaps in compensation and senior leadership, and how they affect firm performance, remain important questions for future research.

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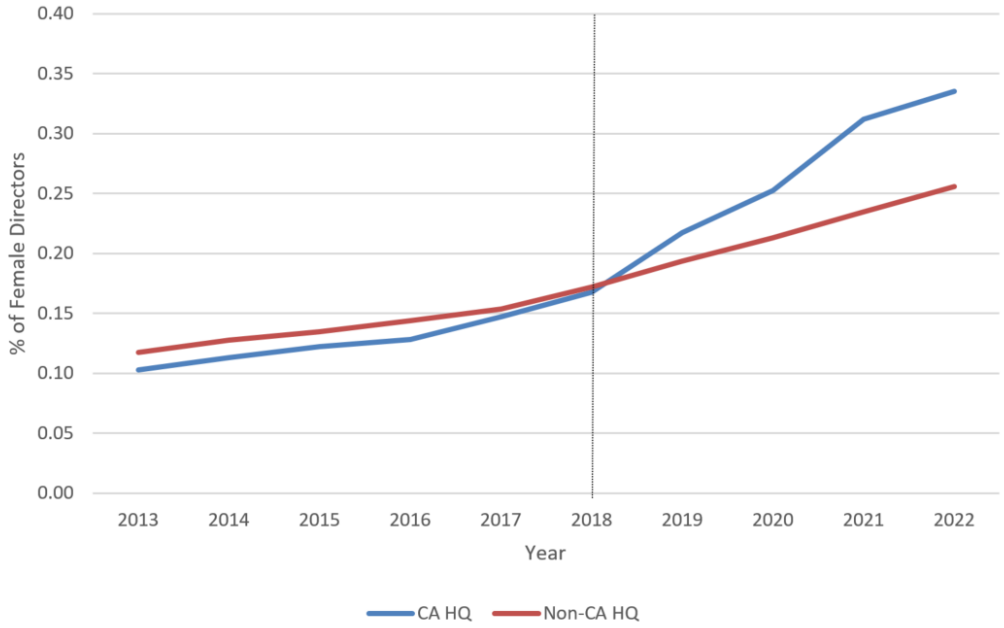
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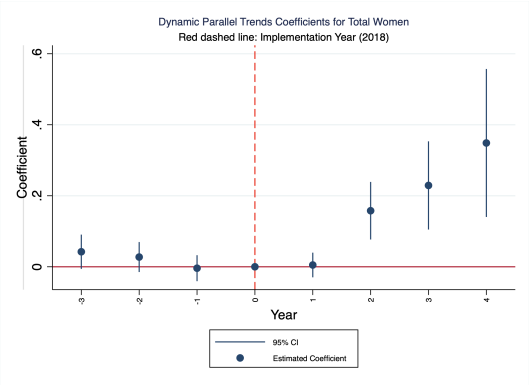
7 Figures

Figure 1. Percent of Board Seats Held By Women

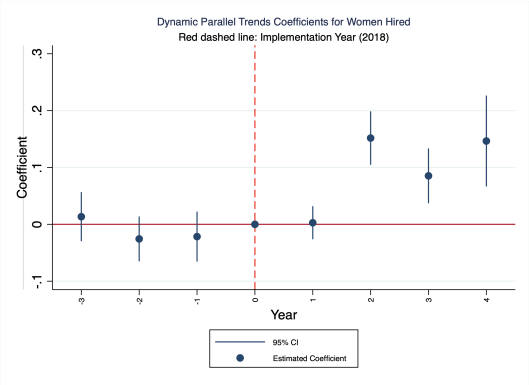


This figure shows the percentage of board seats held by women. The sample consists of public firms in the US covered by BoardEx. Firms headquartered in California are in our treated group while those not headquartered in California are in our control group. The period of observations is between 2013 to 2022. The treatment, passage of the California gender quota, occurred in 2018.

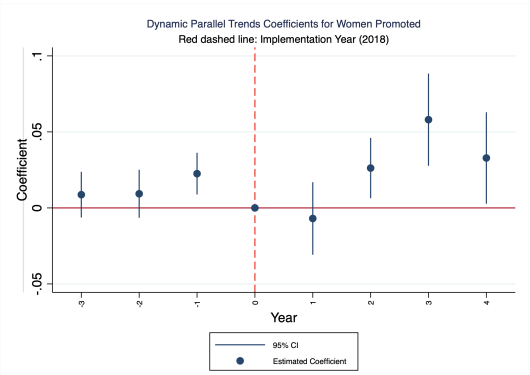
Figure 2. Dynamic Parallel Trends for Women's Career Outcomes



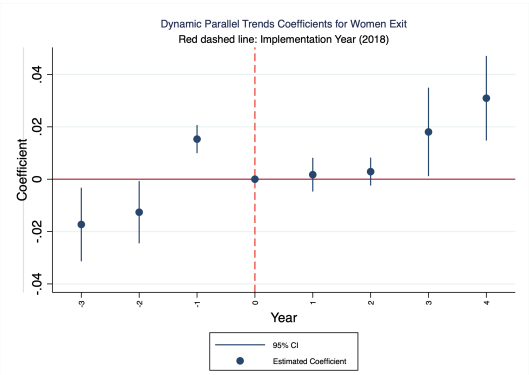
(a) Total Women Normalized by 2017 Women



(b) Hired Women Normalized by 2017 Women



(c) Promoted Women Normalized by 2017 Women Total



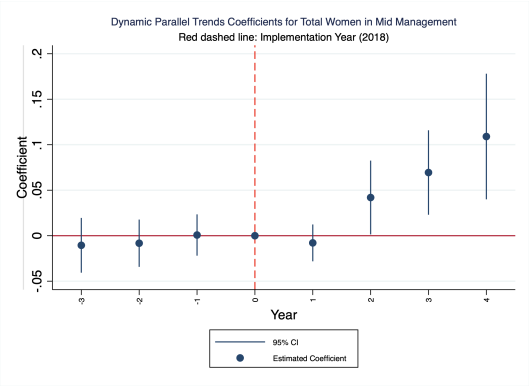
(d) Exiting Women Normalized by 2017 Women Total

This figure presents dynamic parallel trends in career outcomes for women, normalized by the total number of women in 2017. Each subplot represents a different outcome: total employment (top left), hires (top right), promotions (bottom left), and exits (bottom right). The figure plots the estimated coefficients β_k and the 95% confidence interval from the following regression:

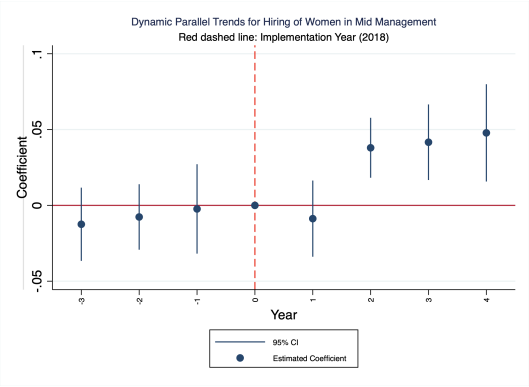
$$Y_{i,t} = \sum_{k=-3, k\neq 0}^{4} \beta_k \cdot \text{Treated}_{i,t}^{(k)} + \alpha_i + \gamma_t + \epsilon_{i,t}$$

where $Y_{i,t}$ represents the outcome variables (total women employed, hired, promoted, and exited, normalized by the total number of women in 2017). The variable $\text{Treated}_{i,t}^{(k)}$ is a set of indicators for each event time k relative to the implementation year (2018), with $k = 0$ excluded as the reference category. The regressions control for firm fixed effects (α_i) and year fixed effects (γ_t), and standard errors are clustered at the state level. The red dashed line marks the policy implementation year (2018). The data includes firm-year observations from 2015 to 2022.

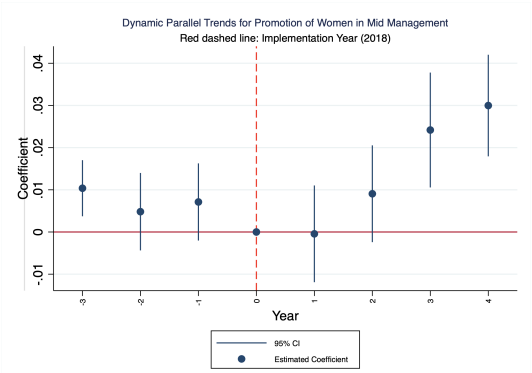
Figure 3. Dynamic Parallel Trends for Women in Mid Management Positions



(a) Total Women in Mid Management Normalized by 2017 Women



(b) Hired Women in Mid Management Normalized by 2017 Women



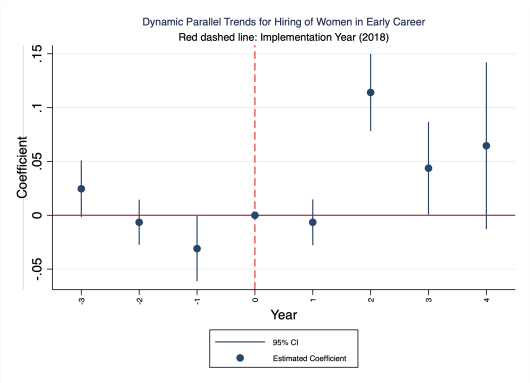
(c) Promoted Women Normalized by 2017 Women

This figure presents dynamic parallel trends in career outcomes for women in mid management positions, normalized by the total number of women in 2017. The left panel shows total women in mid management, and the right panel shows hired women in mid management. The figure plots the estimated coefficients β_k and the 95% confidence intervals from the event-study regression:

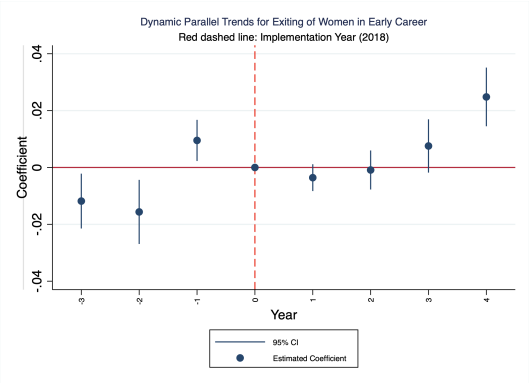
$$Y_{i,t} = \sum_{k=-3, k\neq 0}^{4} \beta_k \cdot \text{Treated}_{i,t}^{(k)} + \alpha_i + \gamma_t + \epsilon_{i,t},$$

where $Y_{i,t}$ is the outcome of interest, and $\text{Treated}_{i,t}^{(k)}$ indicates event time k relative to the 2018 policy. Firm and year fixed effects are included, and standard errors are clustered at the state level. The data covers 2015–2022. The red dashed line marks the policy implementation year (2018).

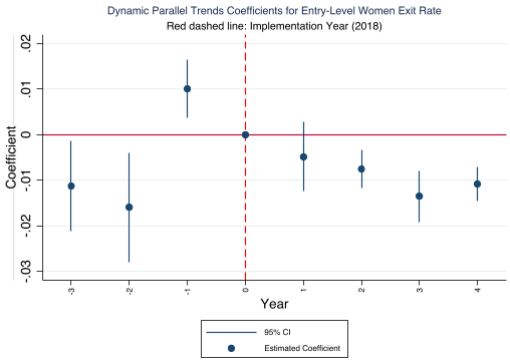
Figure 4. Dynamic Parallel Trends for Women in Early Career Positions



(a) Hired Women in Early Career Normalized by 2017 Women



(b) Exiting Women in Early Career Normalized by 2017 Women



(c) Exiting Rate Women in Early Career Normalized by 2017 Women

This figure presents dynamic parallel trends in career outcomes for women in entry-level positions, normalized by the total number of women in 2017. Each subplot represents a different outcome: hires (top left), exits (top right), and exits rate (bottom). The figure plots the estimated coefficients β_k and the 95% confidence interval from the following regression:

$$Y_{i,t} = \sum_{k=-3, k\neq 0}^{4} \beta_k \cdot \text{Treated}_{i,t}^{(k)} + \alpha_i + \gamma_t + \epsilon_{i,t}$$

where $Y_{i,t}$ represents the outcome variables (total women employed, hired, promoted, and exited, all normalized by the total number of women in 2017). The variable $\text{Treated}_{i,t}^{(k)}$ is a set of indicators for each event time k relative to the implementation year (2018), with $k = 0$ excluded as the reference category. The regressions control for firm fixed effects (α_i) and year fixed effects (γ_t), and standard errors are clustered at the state level. The red dashed line marks the policy implementation year (2018). The data includes firm-year observations from 2015 to 2022.

8 Tables

Table 1
Observable Characteristics in 2017

Panel A: Firm Characteristics

	CA: Mean	Non-CA: Mean	Diff	p-value	CA: No. Obs	Non-CA: No. Obs
Market Value of Equity	3388.22	2100.96	1287.26	0.25	52	205
Book Leverage	0.24	0.26	-0.02	0.68	53	207
PPE	140.16	864.03	-723.87	0.30	49	195
Operating Income	229.40	247.86	-18.46	0.87	49	194
Total Assets	2160.53	3296.26	-1135.73	0.39	54	208
Total Liabilities	1296.26	2296.26	-1000.00	0.31	54	208

Panel B: Overall Board Characteristics

	CA: Mean	Non-CA: Mean	Diff	p-value	CA: No. Obs	Non-CA: No. Obs
Attrition	0.09	0.07	0.02	0.30	19	85
Attrition_3yr	0.12	0.11	0.01	0.40	32	139
Time to Retirement	6.94	5.73	1.21	0.16	47	202
Time in Role	5.93	6.19	-0.26	0.70	47	202
Time on Board	8.05	8.47	-0.42	0.62	47	202
Time In Company	8.39	9.23	-0.84	0.38	47	202
Avg Time in Other Companies	3.38	3.40	-0.02	0.96	47	202
No. of Qualification	2.24	1.97	0.27	0.00	47	202
No. of Directors	6.45	7.31	-0.86	0.00	47	202

Panel C: Sample

	CA: Mean	Non-CA: Mean	Diff	p-value	CA: No. Obs	Non-CA: No. Obs
Top Management %	10%	8%	2%	0.10	54	210
Mid Management %	27%	27%	0%	0.85	54	210
Entry Level %	61%	63%	-2%	0.45	54	210

Table 1 (continued)

Panel D: %Females in Roles

	CA: Mean	Non-CA: Mean	Diff	p-value	CA: No. Obs	Non-CA: No. Obs
Women in Top Management %	15%	19%	-4%	0.36	47	179
Women in Mid Management %	32%	33%	-1%	0.95	52	203
Women in Entry Level %	36%	40%	-4%	0.11	54	208

Panel E: No. Females in Roles

	CA: Mean	Non-CA: Mean	Diff	p-value	CA: No. Obs	Non-CA: No. Obs
Number of Women	52.82	80.91	-28.09	0.31	54	210
Women in Top Management	0.16	0.13	0.03	0.63	54	210
Women in Mid Management	3.65	3.60	0.05	0.9	54	210
Women in Entry Level	38.37	60.3	-21.93	0.31	54	210

Panel F: Hiring, Promotion, and Turn Over

	CA: Mean	Non-CA: Mean	Diff	p-value	CA: No. Obs	Non-CA: No. Obs
Number of Women/Number of Total Employees	0.33	0.36	-0.03	0.26	54	210
Num Women Hired/Num of Employees Hired	0.32	0.36	-0.04	0.38	50	193
Num Women Promoted/Num of Employees Promoted	0.32	0.34	-0.02	0.73	46	164
Num Women Exit /Num of Employees Exit	0.43	0.36	0.07	0.38	29	95

This table reports the means for observable characteristics in our treated(CA firms) and control (Non-CA firms) in the year prior to treatment (2017). Differences are calculated as CA minus Non-CA using the displayed rounded means; differences between percentages are expressed in percentage points. The reported p-values are retained from the original table and have not been recalculated. Panel A reports the financial characteristics of the firm, Panel B reports the board characteristics, Panel C reports the percentage of our sample in the three position levels, Panel D reports the percentage of women in the three position levels, Panel E reports the number of women in each position, and Panel F reports the hiring and turnover activities.

Table 2
Board Gender Composition in the Employment Baseline Sample

	(1) Number of women	(2) Has at least one woman
CA × Post	0.3376***	0.1001***
	(0.0350)	(0.0174)
Firm fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
R^2	0.7549	0.7370
Observations	1,833	1,833
F-statistic	93.18	33.02

Notes: CA × Post is the interaction *CaliforniaHQ* × *Post2018* in Equation (1), where *CaliforniaHQ* indicates a firm headquartered in California in 2017 and *Post2018* indicates years after 2018. This table reports coefficients from firm-panel regressions of board gender composition on CA × Post, with firm and year fixed effects. The dependent variables are the number of female directors on the board (column 1) and an indicator equal to one if the board has at least one female director (column 2). Column 2 is estimated using a linear probability model. The sample includes firm-year observations from 2015–2022 for firms in the employment analysis with available board data, comprising 1,833 observations from 261 firms. Standard errors, clustered at the state level, are reported in parentheses. Board composition data are provided by BoardEx. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 3
Career Progression of Women Across Organizational Levels

	(1) Total	(2) Hired	(3) Promoted	(4) Exit	(5) Exit Rate
Panel A: All Employees					
CA × Post	0.167**	0.103***	0.019**	0.017***	-0.000
	(0.065)	(0.025)	(0.008)	(0.005)	(0.003)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
R^2	0.530	0.387	0.380	0.283	0.269
Observations	2,109	2,109	2,109	2,103	2,115
Panel B: Top Management					
CA × Post	0.056***	0.012***	-0.000	0.003**	0.004***
	(0.005)	(0.002)	(0.001)	(0.001)	(0.001)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
R^2	0.755	0.173	0.157	0.139	0.143
Observations	2,109	2,109	2,109	2,103	2,115
Panel C: Mid-Management					
CA × Post	0.056**	0.034***	0.010***	0.002	0.001
	(0.025)	(0.008)	(0.004)	(0.003)	(0.001)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
R^2	0.624	0.340	0.299	0.168	0.177
Observations	2,109	2,109	2,109	2,103	2,115
Panel D: Entry-Level					
CA × Post	0.053	0.057**	0.004	0.011***	-0.005**
	(0.043)	(0.022)	(0.004)	(0.003)	(0.002)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
R^2	0.526	0.369	0.298	0.289	0.249
Observations	2,109	2,109	2,109	2,103	2,115

CA × Post is the interaction *CaliforniaHQ* × *Post2018* in Equation (1), where *CaliforniaHQ* indicates a firm headquartered in California in 2017 and *Post2018* indicates years after 2018. Columns 1–4 report female employment, hires, promotions, and exits at the indicated organizational level, each divided by the firm's total female workforce in 2017. Column 5 reports female exits divided by contemporaneous female employment at the same organizational level (firm-wide for Panel A). Top management comprises C-suite executives and Vice Presidents, excluding CEOs and Presidents because they may serve on boards. Mid-management comprises employees with "Director" or "Manager" in their position titles; the remaining panel reports entry-level employees. All regressions include firm and year fixed effects. The sample covers 2015–2022. Standard errors clustered at the state level are in parentheses. Employment data are from Bright Data. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4
Career Progression of Women by Industry Gender Composition

	(1) Total	(2) Hired	(3) Promoted	(4) Exit	(5) Exit Rate
Panel A: All Employees					
CA × Post × MD	-0.147	0.005	0.003	-0.081***	-0.030***
	(0.140)	(0.059)	(0.015)	(0.008)	(0.008)
CA × Post	0.292**	0.102*	0.018	0.079***	0.023***
	(0.122)	(0.053)	(0.013)	(0.007)	(0.008)
Post × MD	-0.039	-0.032	-0.013	0.009	0.006
	(0.140)	(0.059)	(0.015)	(0.008)	(0.008)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
R^2	0.537	0.393	0.385	0.289	0.269
Observations	2,097	2,097	2,097	2,097	2,097
Panel B: Top Management Positions					
CA × Post × MD	-0.043***	0.024***	0.012***	-0.003	-0.004**
	(0.014)	(0.005)	(0.002)	(0.002)	(0.002)
CA × Post	0.081***	-0.008	-0.010***	0.006***	0.007***
	(0.012)	(0.005)	(0.001)	(0.001)	(0.001)
Post × MD	-0.034**	-0.005	-0.005***	-0.001	-0.001
	(0.014)	(0.005)	(0.002)	(0.002)	(0.002)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
R^2	0.796	0.186	0.164	0.139	0.144
Observations	2,097	2,097	2,097	2,097	2,097

Table 4 (continued)

	(1) Total	(2) Hired	(3) Promoted	(4) Exit	(5) Exit Rate
Panel C: Mid-Management Positions					
CA × Post × MD	0.122**	0.087***	0.018***	-0.004	0.001
	(0.054)	(0.016)	(0.006)	(0.003)	(0.002)
CA × Post	-0.028	-0.032**	-0.002	0.005***	0.001
	(0.031)	(0.012)	(0.003)	(0.002)	(0.002)
Post × MD	0.008	-0.012	0.005	0.005	0.002
	(0.054)	(0.016)	(0.006)	(0.003)	(0.002)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
R^2	0.638	0.354	0.320	0.169	0.177
Observations	2,097	2,097	2,097	2,097	2,097
Panel D: Entry-Level Positions					
CA × Post × MD	-0.216**	-0.104**	-0.014	-0.074***	-0.027***
	(0.102)	(0.049)	(0.014)	(0.007)	(0.007)
CA × Post	0.231**	0.140***	0.015	0.068***	0.015**
	(0.101)	(0.047)	(0.013)	(0.006)	(0.007)
Post × MD	-0.013	-0.015	-0.005	0.006	0.005
	(0.102)	(0.049)	(0.014)	(0.007)	(0.007)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
R^2	0.532	0.373	0.304	0.298	0.250
Observations	2,097	2,097	2,097	2,097	2,097

CA × Post is the interaction *CaliforniaHQ* × *Post2018* in Equation (1), where *CaliforniaHQ* indicates a firm headquartered in California in 2017 and *Post2018* indicates years after 2018. MD indicates industries in which men accounted for at least 60% of workers in 2017, according to the Bureau of Labor Statistics. Specifications include the fully saturated interactions of CA, Post, and MD, with time-invariant terms absorbed by firm fixed effects. The triple interaction reports the difference in policy effects between male-dominated and other industries. Columns 1–4 report female employment, hires, promotions, and exits at the indicated organizational level, each divided by the firm's total female workforce in 2017. Column 5 reports female exits divided by contemporaneous female employment at the same organizational level (firm-wide for Panel A). Top management comprises C-suite executives and Vice Presidents, excluding CEOs and Presidents because they may serve on boards. Mid-management comprises employees with "Director" or "Manager" in their position titles; the remaining panel reports entry-level employees. All regressions include firm and year fixed effects. The sample covers 2015–2022. Standard errors clustered at the state level are in parentheses. Employment data are from Bright Data. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5
Glassdoor Ratings

	(1) Overall rating	(2) Culture & values	(3) Compen-sation benefits	(4) Career opportu-nities	(5) Work-life balance	(6) Senior leadership
CA × Post	0.276***	0.242***	0.092***	0.170**	0.182***	0.279***
	(0.041)	(0.038)	(0.024)	(0.071)	(0.041)	(0.039)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.410	0.421	0.440	0.361	0.395	0.395
Observations	1,398	1,362	1,362	966	1,363	1,361

CA × Post is the interaction *CaliforniaHQ* × *Post2018* in Equation (1), where *CaliforniaHQ* indicates a firm headquartered in California in 2017 and *Post2018* indicates years after 2018. This table reports coefficients from firm-panel regressions of Glassdoor ratings on CA × Post, with firm fixed effects and year fixed effects. The dependent variables are ratings for overall employer quality (column 1), culture and values (column 2), compensation and benefits (column 3), career opportunities (column 4), work-life balance (column 5), and senior leadership (column 6). Ratings range from one to five. The sample includes firm-year observations from 2015–2022. Standard errors, clustered at the state level, are reported in parentheses. Ratings data are provided by Glassdoor. * ($p < 0.10$), ** ($p < 0.05$), *** ($p < 0.01$).

Table 6
Career Progression and Employee Ratings by Female Board Committee Participation

	(1) Total	(2) Hired	(3) Promoted	(4) Exit	(5) Exit Rate	(6) Overall rating
Panel A: All Employees						
CA × Post × Committee	0.404*	0.166**	0.085***	0.026**	0.011**	0.148*
	(0.217)	(0.074)	(0.030)	(0.013)	(0.005)	(0.081)
CA × Post	-0.035	0.008	-0.032	0.001	-0.008***	0.175**
	(0.202)	(0.072)	(0.024)	(0.014)	(0.003)	(0.083)
Post × Committee	-0.213	-0.090	-0.005	-0.018	-0.010**	-0.018
	(0.217)	(0.074)	(0.030)	(0.013)	(0.005)	(0.080)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.542	0.389	0.379	0.315	0.326	0.449
Observations	1,798	1,798	1,798	1,793	1,807	1,176
Panel B: Top Management Positions						
CA × Post × Committee	0.015	0.035***	0.003	-0.003***	-0.001**	—
	(0.011)	(0.004)	(0.002)	(0.001)	(0.001)	—
CA × Post	0.059***	-0.009***	-0.002	0.004***	0.002***	—
	(0.009)	(0.003)	(0.001)	(0.001)	(0.000)	—
Post × Committee	0.007	-0.003	-0.001	-0.001	-0.001	—
	(0.011)	(0.004)	(0.002)	(0.001)	(0.001)	—
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	—
Year fixed effects	Yes	Yes	Yes	Yes	Yes	—
R^2	0.803	0.186	0.160	0.149	0.145	—
Observations	1,789	1,789	1,789	1,789	1,789	—

Table 6 (continued)

	(1) Total	(2) Hired	(3) Promoted	(4) Exit	(5) Exit Rate	(6) Overall rating
Panel C: Mid-Management Positions						
CA × Post × Committee	0.194**	0.052***	0.041***	0.009	0.003	—
	(0.090)	(0.016)	(0.013)	(0.009)	(0.002)	—
CA × Post	-0.056	0.003	-0.013	-0.004	-0.000	—
	(0.083)	(0.015)	(0.010)	(0.009)	(0.002)	—
Post × Committee	-0.106	-0.018	-0.004	-0.013	-0.006**	—
	(0.090)	(0.016)	(0.013)	(0.009)	(0.002)	—
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	—
Year fixed effects	Yes	Yes	Yes	Yes	Yes	—
R^2	0.635	0.359	0.321	0.185	0.245	—
Observations	1,789	1,789	1,789	1,789	1,789	—
Panel D: Entry-Level Positions						
CA × Post × Committee	0.225*	0.084	0.010	0.020***	0.010**	—
	(0.129)	(0.062)	(0.013)	(0.005)	(0.004)	—
CA × Post	-0.049	0.013	-0.000	0.000	-0.010***	—
	(0.124)	(0.061)	(0.009)	(0.006)	(0.003)	—
Post × Committee	-0.121	-0.070	0.008	-0.004	-0.004	—
	(0.130)	(0.062)	(0.013)	(0.005)	(0.004)	—
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	—
Year fixed effects	Yes	Yes	Yes	Yes	Yes	—
R^2	0.542	0.375	0.300	0.302	0.261	—
Observations	1,789	1,789	1,789	1,789	1,789	—

CA × Post is the interaction *CaliforniaHQ* × *Post2018* in Equation (1), where *CaliforniaHQ* indicates a firm headquartered in California in 2017 and *Post2018* indicates years after 2018, and *Committee* indicates *Committee2019*, a firm with a woman serving on a board committee in 2019. The triple interaction reports how the policy effect differs with female committee participation. Column 6 in Panel A reports the overall Glassdoor rating on a one-to-five scale; dashes in that column in Panels B–D indicate that career-level Glassdoor regressions are not reported. Columns 1–4 report female employment, hires, promotions, and exits at the indicated organizational level, each divided by the firm's total female workforce in 2017. Column 5 reports female exits divided by contemporaneous female employment at the same organizational level (firm-wide for Panel A). Top management comprises C-suite executives and Vice Presidents, excluding CEOs and Presidents because they may serve on boards. Mid-management comprises employees with "Director" or "Manager" in their position titles; the remaining panel reports entry-level employees. All regressions include firm and year fixed effects. The sample covers 2015–2022. Standard errors clustered at the state level are in parentheses. Employment data are from Bright Data. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7
Robustness of Female Employment and Career Outcomes

	(1) Total	(2) Hired	(3) Promoted	(4) Exit	(5) Exit Rate
Panel A: Top 20 Women-Friendly States					
CA × Post	0.187**	0.129***	0.020	0.021***	0.006
	(0.085)	(0.027)	(0.013)	(0.004)	(0.006)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
R^2	0.554	0.401	0.414	0.333	0.304
Observations	1,125	1,125	1,125	1,125	1,125
Panel B: Profit-and-Loss Roles					
CA × Post	0.028***	0.011***	0.006***	-0.002	-0.000
	(0.010)	(0.002)	(0.002)	(0.002)	(0.001)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
R^2	0.723	0.328	0.320	0.162	0.217
Observations	2,097	2,097	2,097	2,097	2,097
Panel C: Controls with at Least 10% of Employees in California					
CA × Post	0.220**	0.150***	0.013	0.024***	0.002
	(0.092)	(0.035)	(0.019)	(0.006)	(0.009)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
R^2	0.539	0.353	0.449	0.292	0.283
Observations	764	764	764	764	764
Panel D: Treated Firms with at Least 20% of Employees Outside California					
CA × Post	0.163**	0.059**	0.033***	-0.008	-0.012***
	(0.065)	(0.026)	(0.008)	(0.005)	(0.003)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
R^2	0.550	0.428	0.388	0.271	0.256
Observations	1,915	1,915	1,915	1,915	1,915
Panel E: Excluding Technology Sectors					
CA × Post	0.302***	0.113***	0.038***	0.030***	0.004
	(0.081)	(0.030)	(0.010)	(0.006)	(0.004)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
R^2	0.544	0.414	0.366	0.285	0.267
Observations	1,788	1,788	1,788	1,788	1,788

Table 7 (continued)

	(1) Total	(2) Hired	(3) Promoted	(4) Exit	(5) Exit Rate
Panel F: Financial controls					
CA × Post	0.3374***	0.1717***	0.0510***	0.0154**	−0.0055*
	(0.0862)	(0.0277)	(0.0098)	(0.0068)	(0.0027)
R^2	0.601	0.450	0.467	0.358	0.376
Observations	1,579	1,579	1,579	1,579	1,579
Panel G: Controls and SIC2 × year FE					
CA × Post	0.3906***	0.2081***	0.0539***	0.0073	−0.0098
	(0.0965)	(0.0305)	(0.0131)	(0.0110)	(0.0060)
R^2	0.667	0.534	0.563	0.451	0.437
Observations	1,412	1,412	1,412	1,412	1,412
Panel H: Industry clustering					
CA × Post	0.3906**	0.2081***	0.0539***	0.0073	−0.0098
	(0.1785)	(0.0687)	(0.0111)	(0.0284)	(0.0167)
R^2	0.667	0.534	0.563	0.451	0.437
Observations	1,412	1,412	1,412	1,412	1,412
Panel I: State HC3 (no Hansen)					
CA × Post	0.3906***	0.2081***	0.0539***	0.0073	−0.0098
	(0.1083)	(0.0340)	(0.0153)	(0.0123)	(0.0066)
R^2	0.667	0.534	0.563	0.451	0.437
Observations	1,412	1,412	1,412	1,412	1,412
Panel J: State wild bootstrap					
CA × Post	0.3906*	0.2081**	0.0539*	0.0073	−0.0098
	—	—	—	—	—
R^2	0.667	0.534	0.563	0.451	0.437
Observations	1,412	1,412	1,412	1,412	1,412

CA × Post is the interaction *CaliforniaHQ* × *Post2018* in Equation (1), where *CaliforniaHQ* indicates a firm headquartered in California in 2017 and *Post2018* indicates years after 2018. This table reports robustness tests for the effects of the board gender quota on women's career outcomes. Each specification reports the coefficient on CA × Post. Columns 1–4 report female employment, hires, promotions, and exits, each divided by the firm's total female workforce in 2017. Column 5 reports female exits divided by contemporaneous female employment. All specifications include firm fixed effects.

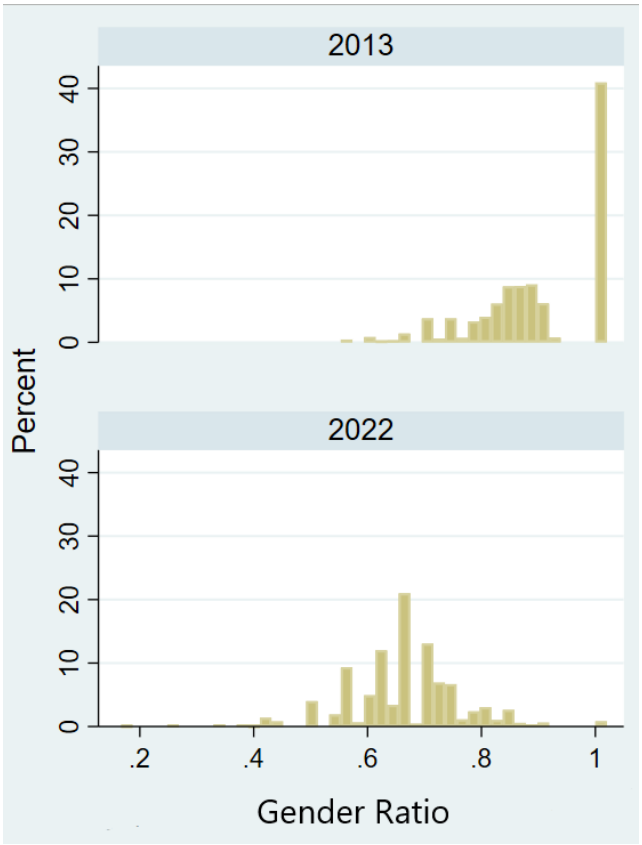
Panel A restricts controls to firms in Oxfam's top 20 states for working women, excluding California. Panel B restricts outcomes to P&L roles, identified by titles indicating general management, operations, logistics, product management, sales, or strategic planning; denominators remain all female employees, as defined above. Panel C restricts controls to firms with at least 10% of employees in California. Panel D restricts treated firms to those with at least 20% of employees outside California in 2017. Panel E excludes BoardEx's Software and Computer Services, Information Technology, and Hardware sectors. Panels A–E include year fixed effects and state-clustered standard errors.

Panels F–J add contemporaneous controls for cash holdings/assets, tangibility, Tobin's Q, book leverage, operating profitability/assets, and capital expenditures/assets; firm size is excluded. Panel F includes year fixed effects, while Panels G–J include two-digit SIC industry-by-year fixed effects. Panels F–G use state clustering; Panel H uses SIC industry clustering; Panel I uses state-clustered HC3 without Hansen or degrees-of-freedom adjustment; and Panel J uses two-sided, null-imposed wild cluster bootstrap-t inference with state-level Rademacher weights and 10,000 repetitions. Panels G–J share the same fitted model; missing SIC data and singleton exclusions reduce their sample.

The sample covers 2015–2022. Employment data are from Bright Data and financial controls from Compustat. Standard errors are in parentheses; dashes indicate unreported statistics. California is the sole treated state, which limits inference. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

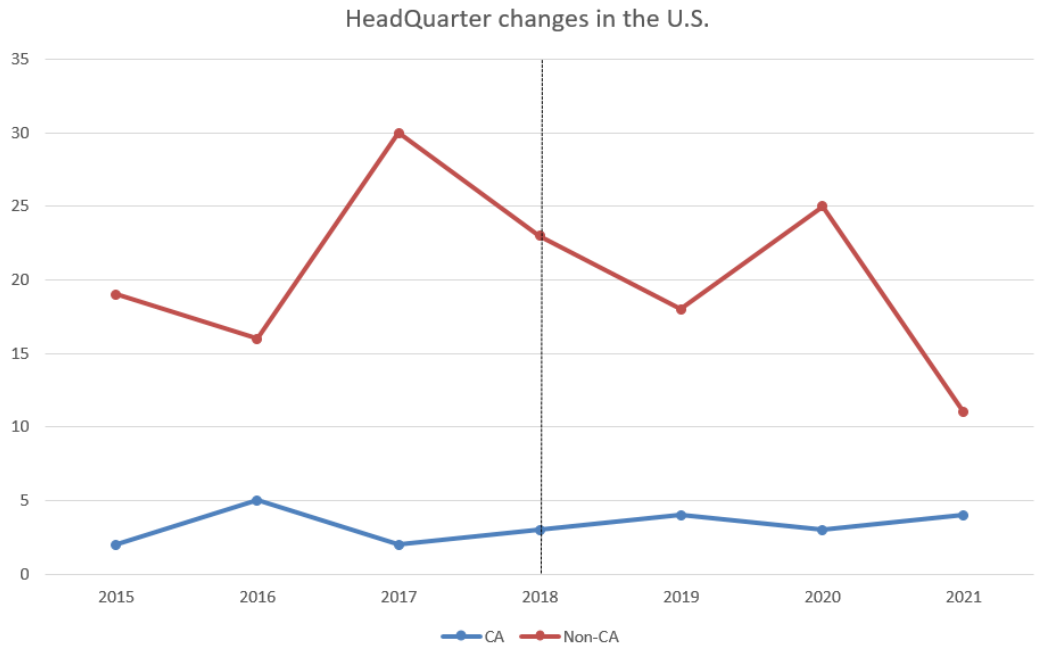
A Supplementary Figures and Tables

Figure A1. Gender Ratio Distribution in California



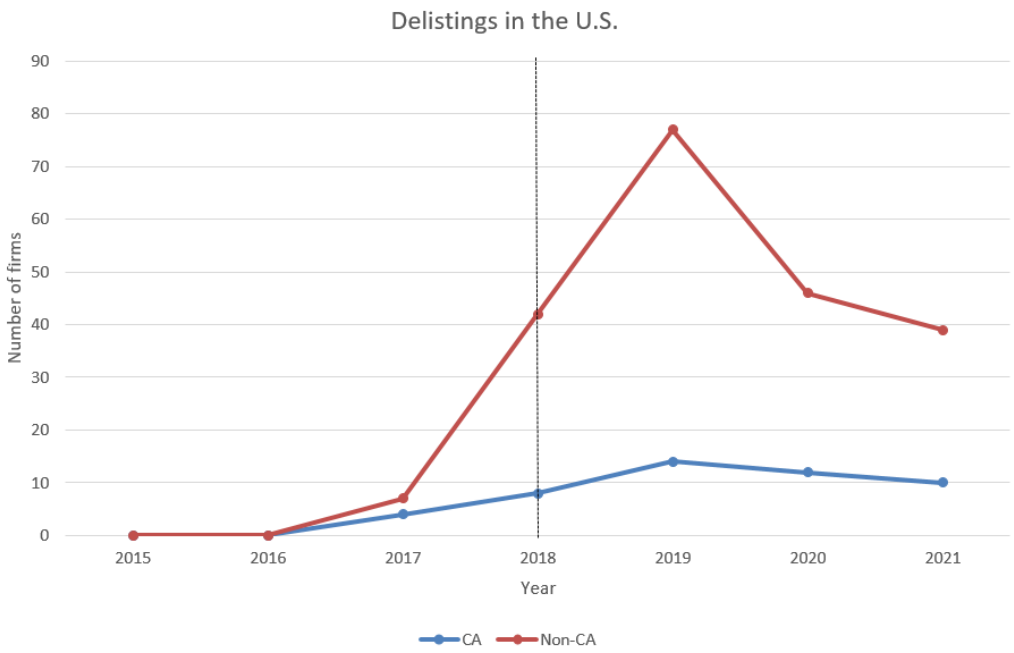
This figure shows the distribution of board gender ratios of public firms in California (treated group) in 2013 and 2022. A gender ratio of 1 represents an all-male board, while a gender ratio of 0 represents an all-female board.

Figure A2. Headquarter Changes in the US



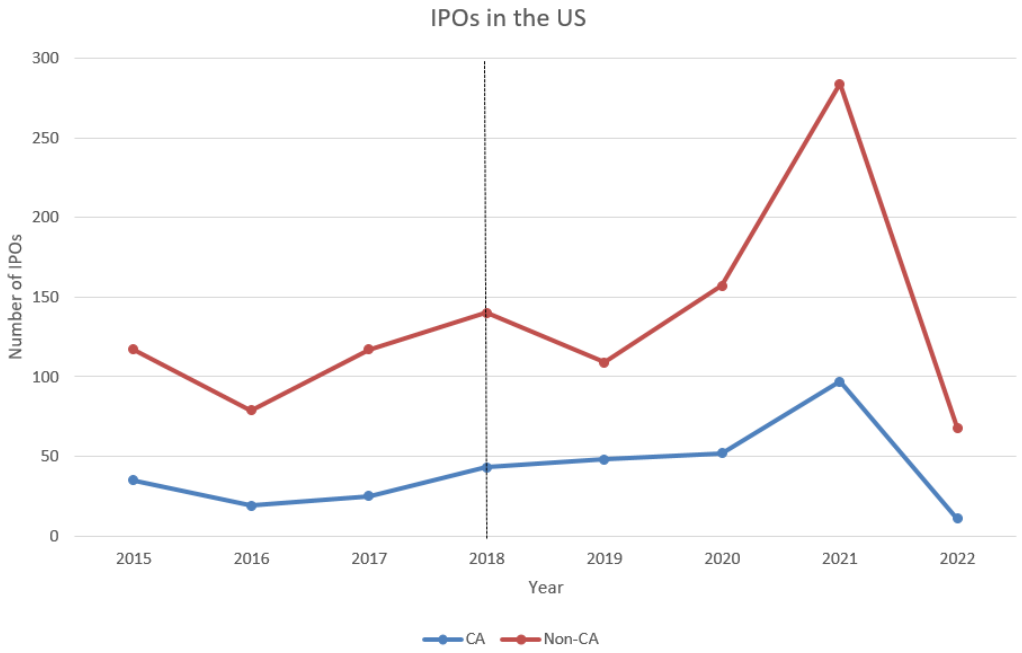
This figure shows the number of firms that changed their headquarters. The sample consists of public firms in the US covered by Compustat. Firms headquartered in California are in our treated group, while those not headquartered in California are in our control group. The period of observations is between 2013 and 2022. The treatment, passage of the California gender quota, occurred in 2018.

Figure A3. Delistings in the US



This figure shows the number of firms that delisted in the US. The sample consists of public firms in the US covered by Compustat. Firms headquartered in California are in our treated group, while those not headquartered in California are in our control group. The period of observations is between 2013 and 2022. The treatment, passage of the California gender quota, occurred in 2018.

Figure A4. IPOs in the US



This figure shows the number of IPOs in the US. The sample consists of firms in the US covered by Compustat. Firms headquartered in California are in our treated group, while those not headquartered in California are in our control group. The period of observations is between 2013 and 2022. The treatment, passage of the California gender quota, occurred in 2018.

Table A1
Gender Quotas Across the World

Country	Quota	Date Law Passed	Date To Comply	Sanction For Non-Compliance
Norway	40% women on the board of PLCs	2003	2007	Forced liquidation
Spain	40% women on the board of PLCs	2007	2015	No consequence for non-compliance.[§]
Iceland	40% women on the board of PLCs with 50+ employees	2010	2013	New firms must comply to go public
Italy	33% women on the board of PLCs	2011	2012	A fine of €100 thousand–€1 million
France	40% women on the board of PLCs	2011	2017/2020[†]	No compensation for directors
Belgium	33% women on the board of all firms	2011	2017/2019[‡]	No compensation for directors
India	1 woman on the board of PLCs[*]	2013	2015	Fine of 50,000–500,000 INR
Germany	30% men and 30% women on the board of PLCs	2015	2016	Fine of €50,000
Austria	30% men and 30% women on the board of PLCs with 1000+ employees	2017	2018	No consequence for non-compliance
Portugal	20% women on the board of public, state-owned, and local government-owned firms	2017	2018	Fine equal to 1 month board compensation

Notes: [*] Listed companies with share capital up to 100 million INR or net worth up to 250 million INR, and small or mid-cap firms are exempt. [†] 2017 for publicly listed firms with more than 500 employees or €50 million in revenue; 2020 for publicly listed firms with more than 250 employees or €50 million in revenue. [‡] 2017 for large-cap publicly listed firms; 2019 for SMEs. [§] Complying firms may be favored for government contracts.

Table A2
Summary Statistics and Mean Comparisons

	CA: Mean	Non-CA: Mean	Diff	p-value	CA: No. Obs	Non-CA: No. Obs
Panel A: Overall						
Women Hired / Women Employed	0.15	0.14	0.01	0.71	54	210
Women Promoted / Women Employed	0.05	0.04	0.01	0.30	54	210
Women Exit / Women Employed	0.05	0.01	0.04	0.00	54	210
Panel B: Top Management						
Women in Top Management	0.08	0.04	0.03	0.13	54	210
Women Hired / Women Employed	0.02	0.00	0.02	0.06	54	210
Women Promoted / Women Employed	0.00	0.00	0.00	0.70	54	210
Women Exit / Women Employed	0.00	0.00	0.00	0.12	54	210
Panel C: Mid-Management						
Women in Mid-Management	0.26	0.24	0.02	0.58	54	210
Women Hired / Women Employed	0.04	0.03	0.01	0.68	54	210
Women Promoted / Women Employed	0.01	0.01	0.00	0.91	54	210
Women Exit / Women Employed	0.01	0.00	0.01	0.18	54	210
Panel D: Entry-Level						
Women in Entry-Level	0.66	0.70	-0.04	0.20	54	210
Women Hired / Women Employed	0.09	0.10	-0.01	0.51	54	210
Women Promoted / Women Employed	0.02	0.02	-0.00	0.91	54	210
Women Exit / Women Employed	0.04	0.01	0.03	0.00	54	210

This table reports the means of the outcome variables for our treated (CA firms) and control (Non-CA firms) groups in the year prior to treatment (2017), along with the difference in means and p-values. The sample consists of 54 California-headquartered firms and 210 firms headquartered outside California, all with all-male boards in 2017. Women in each organizational level, hires, promotions, and exits are each divided by the firm's total female workforce in 2017. Differences are calculated as CA minus Non-CA. The p-values are from two-sided t-tests of equal means. Panel A reports the overall outcome variables; Panel B reports the outcome variables for top management positions; Panel C reports the outcome variables for mid-management roles; Panel D reports the outcome variables for entry-level roles. Employment data are from Bright Data.

Table A3
Career Progression of Women Across Organizational Levels (Male-Dominated Industries)

	(1) Total	(2) Hired	(3) Promoted	(4) Exit	(5) Exit Rate
Panel A: All Employees					
CA × Post	0.145*	0.107***	0.021**	-0.002	-0.007**
	(0.074)	(0.028)	(0.009)	(0.006)	(0.004)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
R^2	0.532	0.388	0.421	0.282	0.281
Observations	1,637	1,637	1,637	1,637	1,637
Panel B: Top Management Positions					
CA × Post	0.038***	0.016***	0.002***	0.003	0.003
	(0.005)	(0.002)	(0.001)	(0.002)	(0.002)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
R^2	0.706	0.144	0.170	0.129	0.129
Observations	1,637	1,637	1,637	1,637	1,637
Panel C: Mid-Management Positions					
CA × Post	0.093***	0.055***	0.015***	0.002	0.001
	(0.034)	(0.008)	(0.004)	(0.003)	(0.002)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
R^2	0.632	0.361	0.317	0.169	0.179
Observations	1,637	1,637	1,637	1,637	1,637
Panel D: Entry-Level Positions					
CA × Post	0.015	0.036	0.001	-0.006*	-0.011***
	(0.044)	(0.023)	(0.004)	(0.003)	(0.002)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
R^2	0.526	0.370	0.338	0.301	0.280
Observations	1,637	1,637	1,637	1,637	1,637

CA × Post is the interaction *CaliforniaHQ* × *Post2018* in Equation (1), where *CaliforniaHQ* indicates a firm headquartered in California in 2017 and *Post2018* indicates years after 2018. This table reports coefficients from firm-panel regressions of the career progression of women on CA × Post, with firm fixed effects and year fixed effects. Panel A reports aggregate outcomes across all organizational levels. Panel B reports outcomes for women in top management positions, defined as C-suite executives and Vice Presidents; CEOs and Presidents are excluded as they may serve on the board of directors. Panel C reports outcomes for women in mid-management positions, defined as employees with "Director" or "Manager" in their position title. Panel D reports outcomes for women in entry-level positions. The dependent variables are the number of women in the relevant category (column 1), the number hired (column 2), the number promoted (column 3), and the number exiting (column 4), each scaled by the firm's total number of female employees in 2017; and the exit rate, defined as the number of women exiting the relevant category divided by the number of women employed in that category in the same year (column 5); Panel A uses all contemporaneous female employees as the denominator. The sample includes firm-year observations from 2015–2022 and is restricted to firms in male-dominated industries, defined as industries in which men accounted for at least 60% of workers in 2017, according to the Bureau of Labor Statistics. Standard errors, clustered at the state level, are reported in parentheses. Our data are provided by Bright Data. * ($p < 0.10$), ** ($p < 0.05$), *** ($p < 0.01$).

Table A4
Career Progression of Women Across Organizational Levels (Non-Male-Dominated Industries)

	(1) Total	(2) Hired	(3) Promoted	(4) Exit	(5) Exit Rate
Panel A: All Employees					
CA × Post	0.292**	0.102*	0.018	0.079***	0.023***
	(0.124)	(0.053)	(0.013)	(0.007)	(0.008)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
R^2	0.558	0.421	0.282	0.313	0.246
Observations	460	460	460	460	460
Panel B: Top Management Positions					
CA × Post	0.081***	-0.008	-0.010***	0.006***	0.007***
	(0.012)	(0.005)	(0.002)	(0.001)	(0.001)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
R^2	0.875	0.295	0.147	0.252	0.316
Observations	460	460	460	460	460
Panel C: Mid-Management Positions					
CA × Post	-0.029	-0.032**	-0.002	0.005***	0.001
	(0.031)	(0.012)	(0.003)	(0.002)	(0.002)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
R^2	0.666	0.298	0.338	0.188	0.186
Observations	460	460	460	460	460
Panel D: Entry-Level Positions					
CA × Post	0.231**	0.140***	0.015	0.068***	0.015**
	(0.103)	(0.048)	(0.013)	(0.006)	(0.007)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
R^2	0.545	0.389	0.232	0.298	0.195
Observations	460	460	460	460	460

CA × Post is the interaction *CaliforniaHQ* × *Post2018* in Equation (1), where *CaliforniaHQ* indicates a firm headquartered in California in 2017 and *Post2018* indicates years after 2018. This table reports coefficients from firm-panel regressions of the career progression of women on CA × Post, with firm fixed effects and year fixed effects. Panel A reports aggregate outcomes across all organizational levels. Panel B reports outcomes for women in top management positions, defined as C-suite executives and Vice Presidents; CEOs and Presidents are excluded as they may serve on the board of directors. Panel C reports outcomes for women in mid-management positions, defined as employees with "Director" or "Manager" in their position title. Panel D reports outcomes for women in entry-level positions. The dependent variables are the number of women in the relevant category (column 1), the number hired (column 2), the number promoted (column 3), and the number exiting (column 4), each scaled by the firm's total number of female employees in 2017; and the exit rate, defined as the number of women exiting the relevant category divided by the number of women employed in that category in the same year (column 5); Panel A uses all contemporaneous female employees as the denominator. The sample includes firm-year observations from 2015–2022 and is restricted to firms in non-male-dominated industries, defined as industries in which men accounted for less than 60% of workers in 2017, according to the Bureau of Labor Statistics. Standard errors, clustered at the state level, are reported in parentheses. Our data are provided by Bright Data. * ($p < 0.10$), ** ($p < 0.05$), *** ($p < 0.01$).

Table A5
Total Workforce Headcount and Gender Quotas

	(1) Total	(2) Hired	(3) Promoted	(4) Exit
CA × Post	-30.277	-7.897**	-2.985	0.301
	(21.864)	(3.730)	(2.182)	(0.241)
Firm fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
R^2	0.907	0.831	0.801	0.845
Observations	2,201	2,201	2,201	2,193

CA × Post is the interaction *CaliforniaHQ* × *Post2018* in Equation (1), where *CaliforniaHQ* indicates a firm headquartered in California in 2017 and *Post2018* indicates years after 2018. This table reports coefficients from firm-panel regressions of total workforce headcount on CA × Post, with firm fixed effects and year fixed effects. The dependent variables are total employees (column 1), total hires (column 2), total promotions (column 3), and total exits (column 4). The sample includes firm-year observations from 2015–2022. Standard errors, clustered at the state level, are reported in parentheses. Data are provided by Bright Data. * ($p < 0.10$), ** ($p < 0.05$), *** ($p < 0.01$).

Table A6
Career Progression of Women: Top Management, Mid-Management, and Entry-Level Positions (Top 20 States)

	(1) Total	(2) Hired	(3) Promoted	(4) Exit	(5) Exit Rate
Panel A: Top Management Positions					
CA × Post	0.044***	0.008**	-0.000	0.005	0.005
	(0.008)	(0.003)	(0.001)	(0.003)	(0.003)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
R^2	0.779	0.190	0.178	0.142	0.147
Observations	1,125	1,125	1,125	1,125	1,125
Panel B: Mid-Management Positions					
CA × Post	0.028	0.032**	0.008	0.006**	0.004
	(0.042)	(0.013)	(0.005)	(0.002)	(0.003)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
R^2	0.639	0.351	0.308	0.209	0.211
Observations	1,125	1,125	1,125	1,125	1,125
Panel C: Entry-Level Positions					
CA × Post	0.112**	0.086***	0.008	0.011***	-0.004
	(0.042)	(0.020)	(0.006)	(0.003)	(0.003)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
R^2	0.563	0.389	0.320	0.327	0.298
Observations	1,125	1,125	1,125	1,125	1,125

CA × Post is the interaction *CaliforniaHQ* × *Post2018* in Equation (1), where *CaliforniaHQ* indicates a firm headquartered in California in 2017 and *Post2018* indicates years after 2018. This table reports coefficients from firm-panel regressions of the career progression of women on CA × Post, with firm fixed effects and year fixed effects. Panel A reports outcomes for women in top management positions, defined as C-suite executives and Vice Presidents; CEOs and Presidents are excluded as they may serve on the board of directors. Panel B reports outcomes for women in mid-management positions, defined as employees with "Director" or "Manager" in their position title. Panel C reports outcomes for women in entry-level positions. Results for all employees are reported in Table 7, Panel A. The dependent variables are the number of women in the relevant category (column 1), the number hired (column 2), the number promoted (column 3), and the number exiting (column 4), each scaled by the firm's total number of female employees in 2017; and the exit rate, defined as the number of women exiting the relevant category divided by the number of women employed in that category in the same year (column 5). The control group is restricted to firms headquartered in the top 20 states for working women identified by Oxfam, excluding California. California-headquartered firms remain in the treated group. The sample includes firm-year observations from 2015–2022. Standard errors, clustered at the state level, are reported in parentheses. Our data are provided by Bright Data. * ($p < 0.10$), ** ($p < 0.05$), *** ($p < 0.01$).

Table A7
Career Progression of Women: Top Management, Mid-Management, and Entry-Level Positions (Profit & Loss Roles)

	(1) Total	(2) Hired	(3) Promoted	(4) Exit	(5) Exit Rate
Panel A: Top Management Positions					
CA × Post	0.002	0.000	0.000	-0.000	0.000
	(0.001)	(0.000)	(0.000)	(0.000)	(0.000)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
R^2	0.565	0.160	0.136	0.131	0.133
Observations	2,097	2,097	2,097	2,097	2,097
Panel B: Mid-Management Positions					
CA × Post	0.018**	0.005***	0.001	-0.001	0.000
	(0.009)	(0.002)	(0.001)	(0.002)	(0.000)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
R^2	0.664	0.273	0.214	0.130	0.158
Observations	2,097	2,097	2,097	2,097	2,097
Panel C: Entry-Level Positions					
CA × Post	0.008***	0.006***	0.000	-0.001	-0.000
	(0.003)	(0.001)	(0.001)	(0.001)	(0.000)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
R^2	0.745	0.267	0.236	0.220	0.214
Observations	2,097	2,097	2,097	2,097	2,097

CA × Post is the interaction *CaliforniaHQ* × *Post2018* in Equation (1), where *CaliforniaHQ* indicates a firm headquartered in California in 2017 and *Post2018* indicates years after 2018. This table reports coefficients from firm-panel regressions of the career progression of women on CA × Post, with firm fixed effects and year fixed effects. Panel A reports outcomes for women in top management positions, defined as C-suite executives and Vice Presidents; CEOs and Presidents are excluded as they may serve on the board of directors. Panel B reports outcomes for women in mid-management positions, defined as employees with "Director" or "Manager" in their position title. Panel C reports outcomes for women in entry-level positions. Results for all employees, scaled by the firm's total female workforce in 2017, are reported in Table 7, Panel B; Table A8 reports the same outcomes scaled by female P&L employment. The dependent variables are the number of women in the relevant category (column 1), the number hired (column 2), the number promoted (column 3), and the number exiting (column 4), each scaled by the firm's total number of female employees in 2017; and the exit rate, defined as the number of women exiting P&L roles in the relevant category divided by the total number of women employed in that category in the same year (column 5). Outcomes are restricted to profit-and-loss (P&L) roles, defined as positions with titles indicating general management, operations, logistics, product management, sales, or strategic planning. The sample includes firm-year observations from 2015–2022. Standard errors, clustered at the state level, are reported in parentheses. Our data are provided by Bright Data. * ($p < 0.10$), ** ($p < 0.05$), *** ($p < 0.01$).

Table A8
Career Progression of Women Across Organizational Levels (Profit-and-Loss Roles, P&L Baseline)

	(1) Total	(2) Hired	(3) Promoted	(4) Exit	(5) Exit Rate
Panel A: All Employees					
CA × Post	0.371***	0.076***	0.087***	-0.005	0.008***
	(0.064)	(0.021)	(0.024)	(0.007)	(0.003)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
R^2	0.552	0.359	0.327	0.350	0.240
Observations	1,380	1,380	1,380	1,380	1,380
Panel B: Top Management Positions					
CA × Post	0.020**	-0.003	-0.009***	0.000	0.001***
	(0.010)	(0.003)	(0.003)	(0.000)	(0.000)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
R^2	0.747	0.193	0.143	0.245	0.234
Observations	1,380	1,380	1,380	1,380	1,380
Panel C: Mid-Management Positions					
CA × Post	0.289***	0.018	0.034***	0.001	0.004
	(0.050)	(0.012)	(0.011)	(0.006)	(0.004)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
R^2	0.639	0.251	0.233	0.137	0.149
Observations	1,380	1,380	1,380	1,380	1,380
Panel D: Entry-Level Positions					
CA × Post	0.062**	0.061***	0.027***	-0.007	0.003
	(0.026)	(0.012)	(0.009)	(0.010)	(0.003)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
R^2	0.553	0.304	0.261	0.284	0.221
Observations	1,380	1,380	1,380	1,380	1,380

CA × Post is the interaction *CaliforniaHQ* × *Post2018* in Equation (1), where *CaliforniaHQ* indicates a firm headquartered in California in 2017 and *Post2018* indicates years after 2018. This table reports coefficients from firm-panel regressions of the career progression of women on CA × Post, with firm fixed effects and year fixed effects. Panel A reports aggregate outcomes across all organizational levels. Panel B reports outcomes for women in top management positions, defined as C-suite executives and Vice Presidents; CEOs and Presidents are excluded as they may serve on the board of directors. Panel C reports outcomes for women in mid-management positions, defined as employees with "Director" or "Manager" in their position title. Panel D reports outcomes for women in entry-level positions. The dependent variables are the number of women in the relevant category (column 1), the number hired (column 2), the number promoted (column 3), and the number exiting (column 4), each scaled by the firm's total number of female P&L employees in 2017; and the exit rate, defined as the number of women exiting P&L roles in the relevant category divided by the number of women employed in P&L roles in that category in the same year (column 5); Panel A uses all contemporaneous female P&L employees as the denominator. Outcomes are restricted to profit-and-loss (P&L) roles, defined as positions with titles indicating general management, operations, logistics, product management, sales, or strategic planning. The sample includes firm-year observations from 2015–2022. Standard errors, clustered at the state level, are reported in parentheses. Our data are provided by Bright Data. * ($p < 0.10$), ** ($p < 0.05$), *** ($p < 0.01$).

Table A9
Career Progression of Women by Organizational Level: Control Firms with at Least 10% of Employees in California

	(1) Total	(2) Hired	(3) Promoted	(4) Exit	(5) Exit Rate
Panel A: Top Management Positions					
CA × Post	0.047***	0.009***	-0.002	0.003***	0.003**
	(0.009)	(0.002)	(0.002)	(0.001)	(0.001)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
R^2	0.758	0.195	0.172	0.273	0.286
Observations	764	764	764	764	764
Panel B: Mid-Management Positions					
CA × Post	0.056	0.041**	0.003	0.008**	0.007
	(0.036)	(0.018)	(0.013)	(0.003)	(0.004)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
R^2	0.605	0.321	0.365	0.202	0.191
Observations	764	764	764	764	764
Panel C: Entry-Level Positions					
CA × Post	0.112*	0.101***	0.009	0.014**	-0.008
	(0.057)	(0.020)	(0.007)	(0.005)	(0.008)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
R^2	0.535	0.355	0.445	0.289	0.260
Observations	764	764	764	764	764

CA × Post is the interaction *CaliforniaHQ* × *Post2018* in Equation (1), where *CaliforniaHQ* indicates a firm headquartered in California in 2017 and *Post2018* indicates years after 2018. This table reports coefficients from firm-panel regressions of the career progression of women on CA × Post, with firm fixed effects and year fixed effects. Panel A reports outcomes for women in top management positions, defined as C-suite executives and Vice Presidents; CEOs and Presidents are excluded as they may serve on the board of directors. Panel B reports outcomes for women in mid-management positions, defined as employees with "Director" or "Manager" in their position title. Panel C reports outcomes for women in entry-level positions. Results for all employees are reported in Table 7, Panel C. The dependent variables are the number of women in the relevant category (column 1), the number hired (column 2), the number promoted (column 3), and the number exiting (column 4), each scaled by the firm's total number of female employees in 2017; and the exit rate, defined as the number of women exiting the relevant category divided by the number of women employed in that category in the same year (column 5). The control group is restricted to firms with at least 10% of their employees located in California. The sample includes firm-year observations from 2015–2022. Standard errors, clustered at the state level, are reported in parentheses. Our data are provided by Bright Data. * ($p < 0.10$), ** ($p < 0.05$), *** ($p < 0.01$).

Table A10
Career Progression of Women by Organizational Level: Treated Firms with at Least 20% of Employees Outside California

	(1) Total	(2) Hired	(3) Promoted	(4) Exit	(5) Exit Rate
Panel A: Top Management Positions					
CA × Post	0.032***	0.004*	0.002***	0.001	0.002
	(0.005)	(0.002)	(0.001)	(0.001)	(0.001)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
R^2	0.793	0.175	0.159	0.130	0.129
Observations	1,915	1,915	1,915	1,915	1,915
Panel B: Mid-Management Positions					
CA × Post	0.066**	0.018**	0.008**	0.001	-0.000
	(0.025)	(0.007)	(0.003)	(0.003)	(0.001)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
R^2	0.651	0.364	0.337	0.167	0.176
Observations	1,915	1,915	1,915	1,915	1,915
Panel C: Entry-Level Positions					
CA × Post	0.067	0.038	0.010**	-0.011***	-0.014***
	(0.044)	(0.023)	(0.004)	(0.003)	(0.002)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
R^2	0.541	0.389	0.302	0.276	0.240
Observations	1,915	1,915	1,915	1,915	1,915

CA × Post is the interaction *CaliforniaHQ* × *Post2018* in Equation (1), where *CaliforniaHQ* indicates a firm headquartered in California in 2017 and *Post2018* indicates years after 2018. This table reports coefficients from firm-panel regressions of career progression of women on CA × Post, with firm fixed effects and year fixed effects. Panel A reports outcomes for women in top management positions, defined as C-suite executives and Vice Presidents; CEOs and Presidents are excluded as they may serve on the board of directors. Panel B reports outcomes for women in mid-management positions, defined as employees with "Director" or "Manager" in their position title. Panel C reports outcomes for women in entry-level positions. Results for all employees are reported in Table 7, Panel D. The dependent variables are the number of women in the relevant category (column 1), the number hired (column 2), the number promoted (column 3), and the number exiting (column 4), each scaled by the firm's total number of female employees in 2017; and the exit rate, defined as the number of women exiting the relevant category divided by the number of women employed in that category in the same year (column 5). The treated group is restricted to California-headquartered firms with at least 20% of their workforce located outside California in 2017. The sample includes firm-year observations from 2015–2022. Standard errors, clustered at the state level, are reported in parentheses. Data are provided by Bright Data. * ($p < 0.10$), ** ($p < 0.05$), *** ($p < 0.01$).

Table A11
Career Progression of Women: Top Management, Mid-Management, and Entry-Level Positions (Excluding Technology Sectors)

	(1) Total	(2) Hired	(3) Promoted	(4) Exit	(5) Exit Rate
Panel A: Top Management Positions					
CA × Post	0.048***	0.003	-0.000	0.004**	0.004***
	(0.005)	(0.002)	(0.001)	(0.001)	(0.001)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
R^2	0.840	0.234	0.156	0.140	0.144
Observations	1,788	1,788	1,788	1,788	1,788
Panel B: Mid-Management Positions					
CA × Post	0.063**	0.015**	0.010**	0.005	0.002
	(0.031)	(0.007)	(0.004)	(0.003)	(0.002)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
R^2	0.648	0.358	0.326	0.155	0.158
Observations	1,788	1,788	1,788	1,788	1,788
Panel C: Entry-Level Positions					
CA × Post	0.189***	0.095***	0.018***	0.021***	-0.002
	(0.053)	(0.027)	(0.004)	(0.003)	(0.003)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes
R^2	0.524	0.376	0.279	0.296	0.253
Observations	1,788	1,788	1,788	1,788	1,788

CA × Post is the interaction *CaliforniaHQ* × *Post2018* in Equation (1), where *CaliforniaHQ* indicates a firm headquartered in California in 2017 and *Post2018* indicates years after 2018. This table reports coefficients from firm-panel regressions of the career progression of women on CA × Post, with firm fixed effects and year fixed effects. Panel A reports outcomes for women in top management positions, defined as C-suite executives and Vice Presidents; CEOs and Presidents are excluded as they may serve on the board of directors. Panel B reports outcomes for women in mid-management positions, defined as employees with "Director" or "Manager" in their position title. Panel C reports outcomes for women in entry-level positions. Results for all employees are reported in Table 7, Panel E. The dependent variables are the number of women in the relevant category (column 1), the number hired (column 2), the number promoted (column 3), and the number exiting (column 4), each scaled by the firm's total number of female employees in 2017; and the exit rate, defined as the number of women exiting the relevant category divided by the number of women employed in that category in the same year (column 5). The sample excludes technology firms, defined as firms classified by BoardEx in the "Software and Computer Services," "Information Technology," or "Hardware" sectors. The sample includes firm-year observations from 2015–2022. Standard errors, clustered at the state level, are reported in parentheses. Our data are provided by Bright Data. * ($p < 0.10$), ** ($p < 0.05$), *** ($p < 0.01$).